Integrated Circuit ECE326
Lec. 09
CMOS inverter Characteristics

Dr. Ahmed Reda Mohamed[1,2]

[1]Ph.D, the MoE Key Laboratory of Artificial Intelligence & Bio-Circuits and Systems Laboratory (*BiCASL*), Department of Micro-Nano Electronics, Shanghai Jiao Tong University, Shanghai, China.
[2]Assistant professor, Department of Electronics and Communications, Zagazig University, Zagazig , Egypt.

Outlines

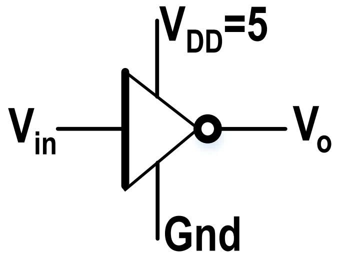
- DC transfer Ch/CS Curve
- Noise Margin Concept
- MOS Inverter Topologies
- DC Transfer Characteristics
- Beta Ratio Effects
- Dynamic Characteristics of CMOS Inverter
- Inverter switching characteristics
- Power Dissipation
- Discussion

Outlines

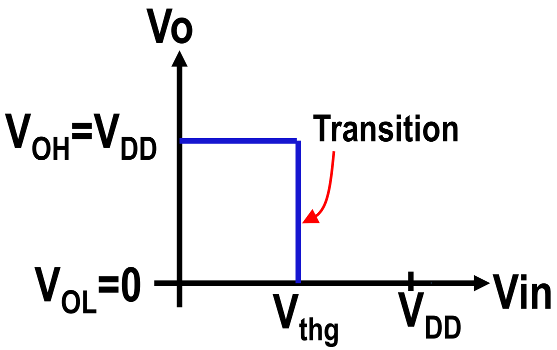
- ❑ DC transfer Ch/CS Curve
- ❑ Noise Margin Concept
- ❑ MOS Inverter Topologies
- ❑ DC Transfer Characteristics
- ❑ Beta Ratio Effects
- ❑ Dynamic Characteristics of CMOS Inverter
- ❑ Inverter switching characteristics
- ❑ Power Dissipation
- ❑ Discussion

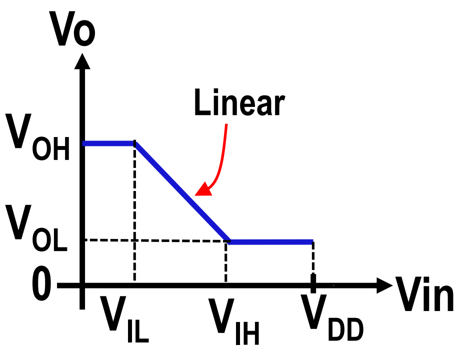
DC transfer Ch/CS curve

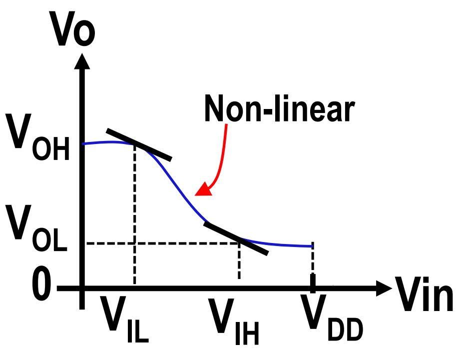
- **Definition**
  - It is a relation between V_{in} and V_o



- **Types of Ch/Cs curve**
  - Ideal curve
  - Simplified curve
  - Practical curve







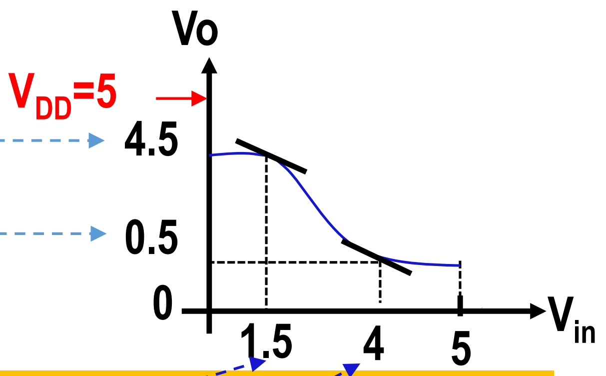
- **Notes**

V_{OH}: Minimum output voltage can be considered as "1"

V_{OL}: Maximum output voltage can be considered as "0"

V_{IL} : Maximum input voltage can be considered as "0"

V_{IH} : Minimum input voltage can be considered as "1"



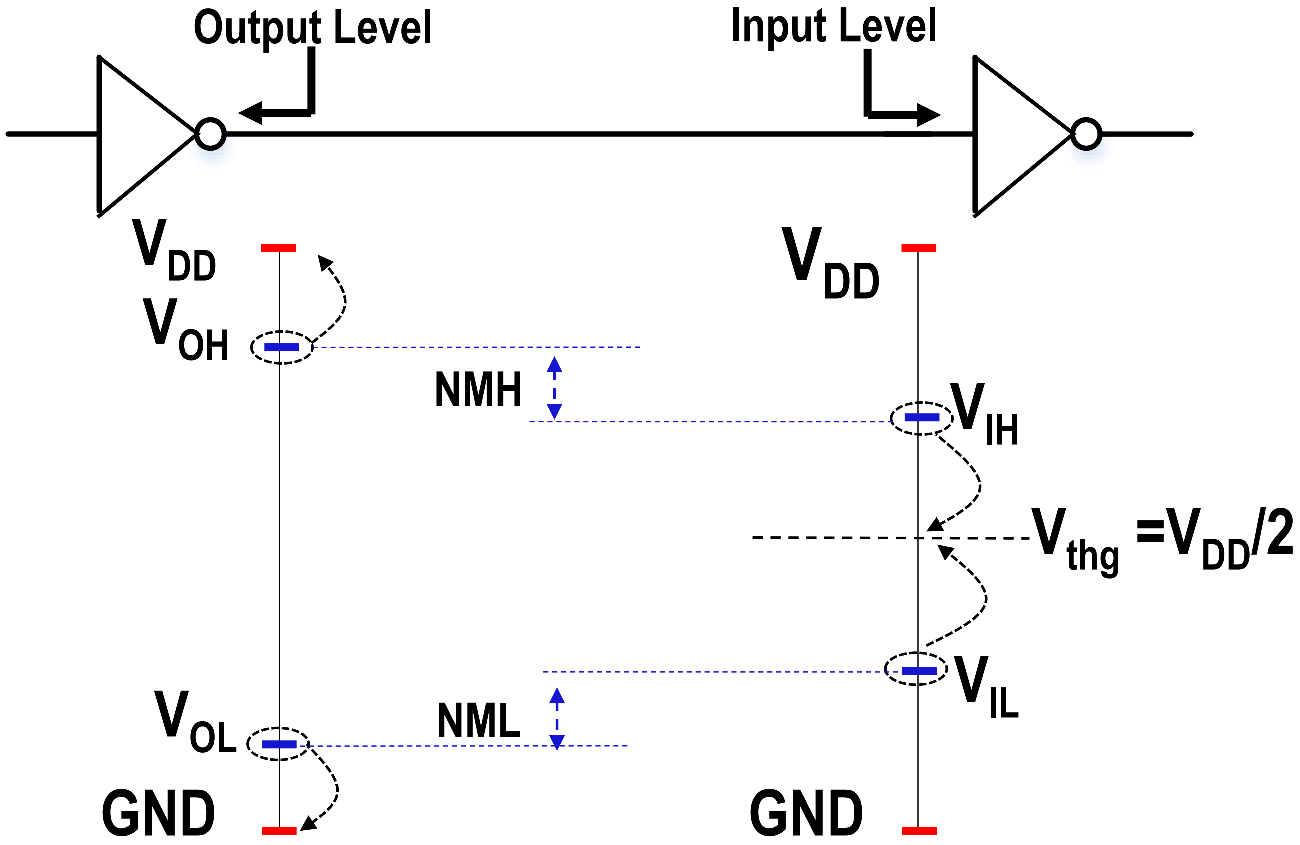
Ideally, $V_{OH}=V_{DD}$, V_{OL}=GND
$V_{IH}=V_{IL}=V_{DD}/2$

Outlines

- ❑ DC transfer Ch/CS Curve
- ❑ Noise Margin Concept
- ❑ MOS Inverter Topologies
- ❑ DC Transfer Characteristics
- ❑ Beta Ratio Effects
- ❑ Dynamic Characteristics of CMOS Inverter
- ❑ Inverter switching characteristics
- ❑ Power Dissipation
- ❑ Discussion

Noise Margin Concept (1/2)

- **Definition**
  - It indicates how system can immune against the noise
- **Categories of Noise Margin**
  - $NMH = V_{OH} - V_{IH}$
  - $NML = V_{IL} - V_{OL}$



Remember

Ideally, $V_{OH}=V_{DD}$, $V_{OL}=GND$

$V_{IH}=V_{IL}=V_{DD}/2$

⬇

$NMH=V_{DD}/2$

$NML=V_{DD}/2$

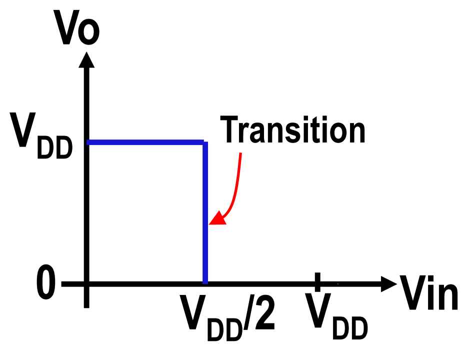
Ideal CMOS inverter

- $NMH = V_{OH} - V_{IH}$
  - $V_{OH} = V_{DD}$, $V_{IH} = V_{DD}/2$
- $NML = V_{IH} - V_{OL}$
  - $V_{IH} = V_{DD}/2$, $V_{OL} = GND$

$\Rightarrow NMH = V_{DD}/2$

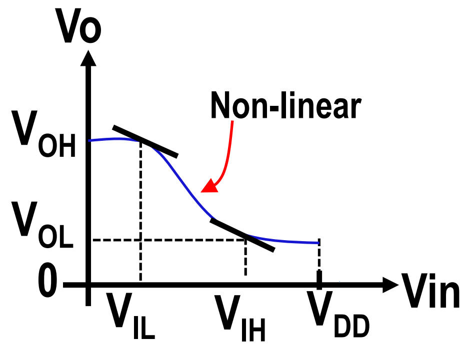
$\Rightarrow NML = V_{DD}/2$

$\Rightarrow NMH = NML$



Practical CMOS inverter

- <u>To get V_{OH}</u>
  - Apply Vin=0, then find $V_o = V_{OH}$
- <u>To get V_{OL}</u>
  - Apply $V_{in} = V_{DD}$, then find $V_o = V_{OL}$
- <u>To get V_{IH} and V_{IL}</u>
  - Find relation between V_o and V_{in}, then substitute $\frac{dV_o}{dV_{in}} = -1.$

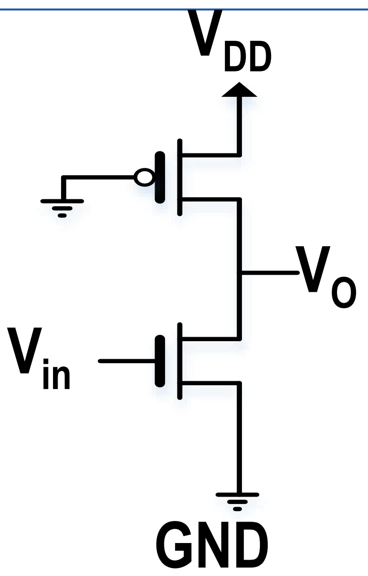


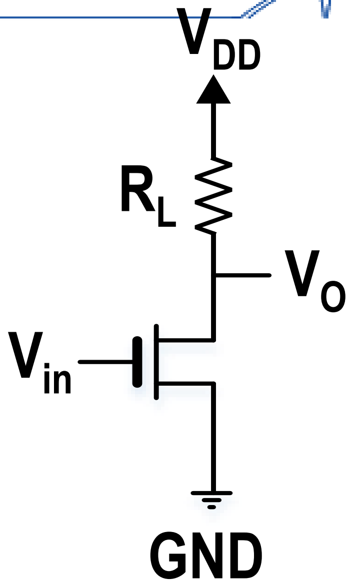
Outlines

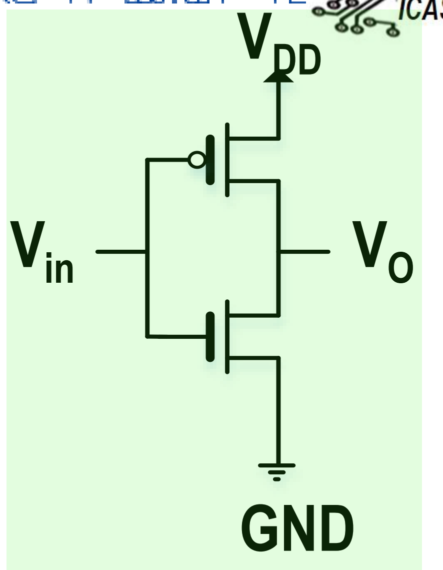
- ❑ DC transfer Ch/CS Curve
- ❑ Noise Margin Concept
- ❑ MOS Inverter Topologies
- ❑ DC Transfer Characteristics
- ❑ Beta Ratio Effects
- ❑ Dynamic Characteristics of CMOS Inverter
- ❑ Inverter switching characteristics
- ❑ Power Dissipation
- ❑ Discussion

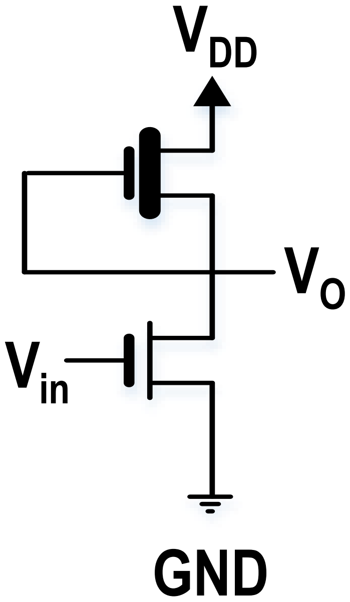
MOS Inverter Topologies

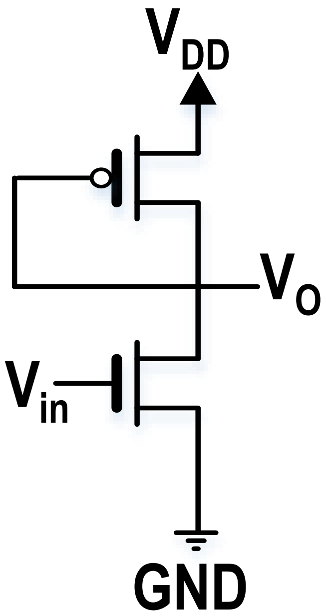
- ❑ **CMOS Inverter**
- ❑ Static Load Inverter
- ❑ Pseudo (N-MOS) Inverter
- ❑ Saturation (N-MOS) Inverter
- ❑ More Saturation (N-MOS) Inverter
- ❑ Depletion (N-MOS) Inverter.

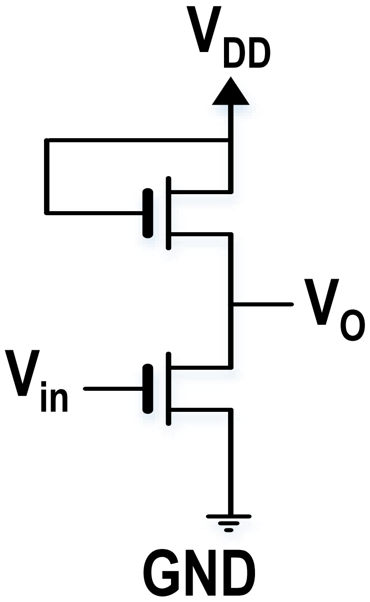










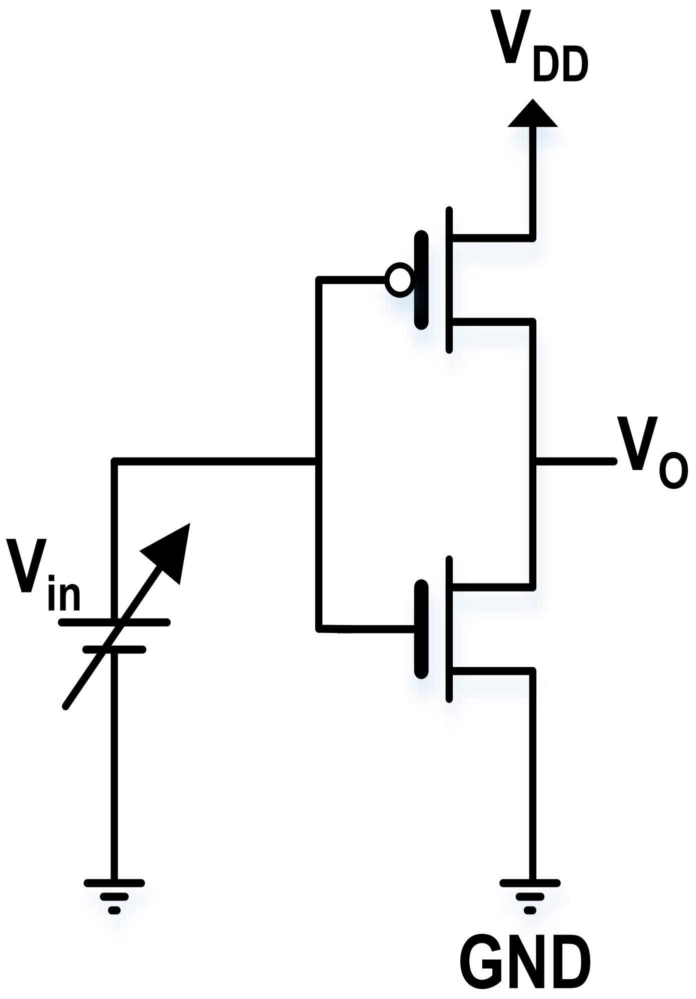


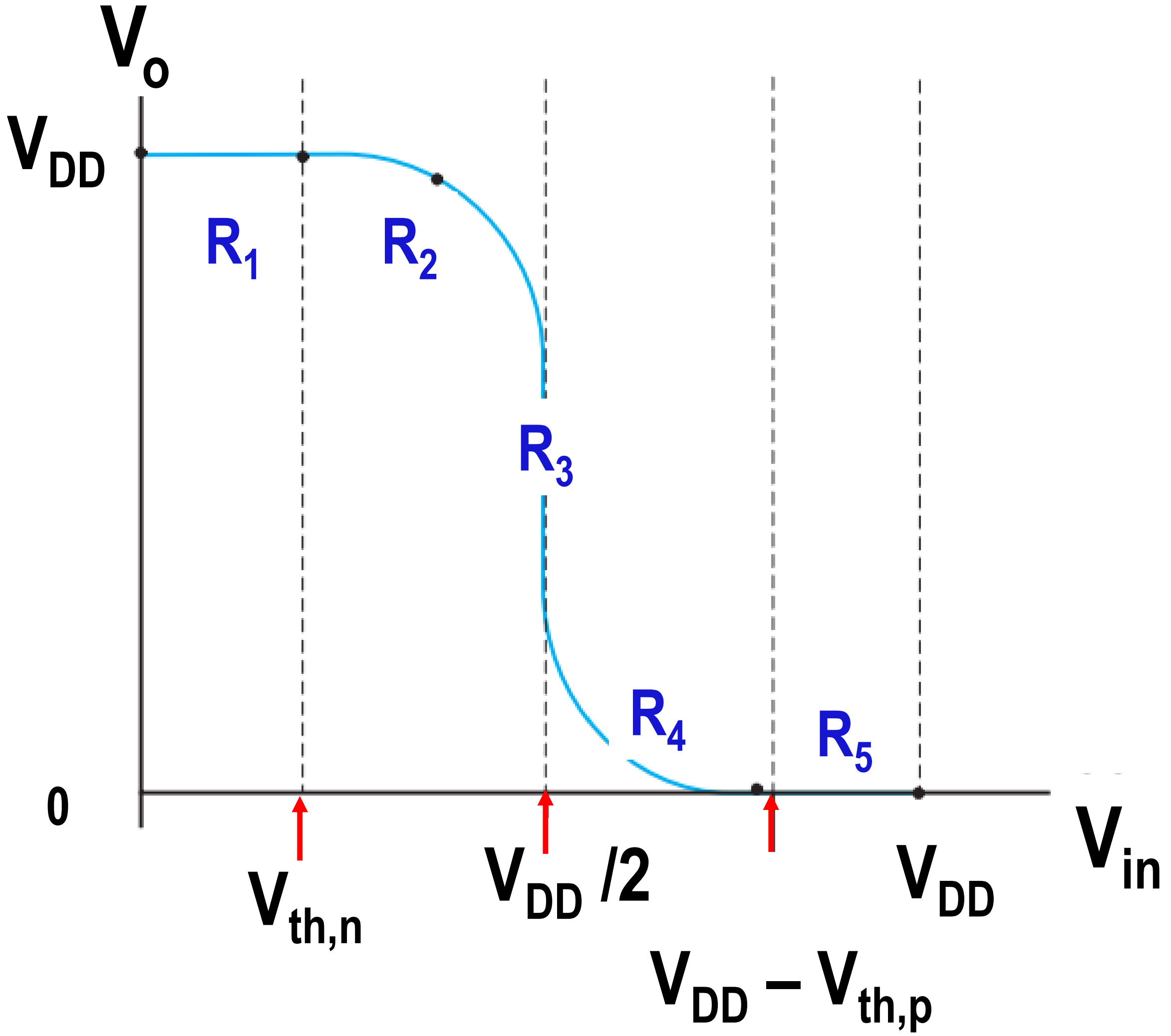
Outlines

- ❑ DC transfer Ch/CS Curve
- ❑ Noise Margin Concept
- ❑ MOS Inverter Topologies
- ❑ DC Transfer Characteristics
- ❑ Beta Ratio Effects
- ❑ Dynamic Characteristics of CMOS Inverter
- ❑ Inverter switching characteristics
- ❑ Power Dissipation
- ❑ Discussion

DC Transfer Characteristics (1/8)

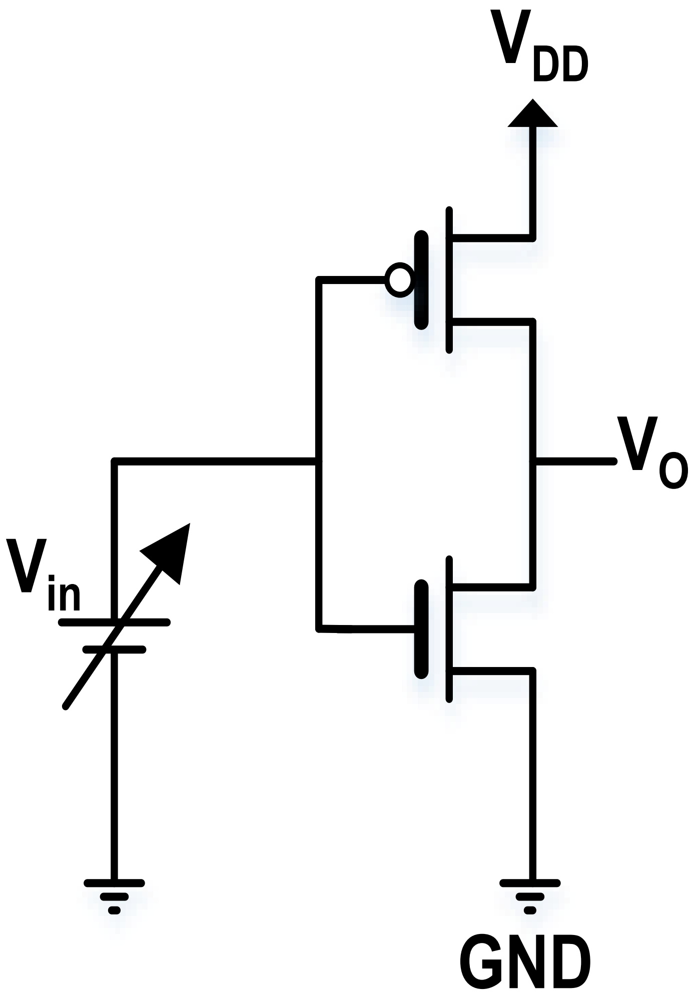
- Static CMOS Inverter DC Characteristics

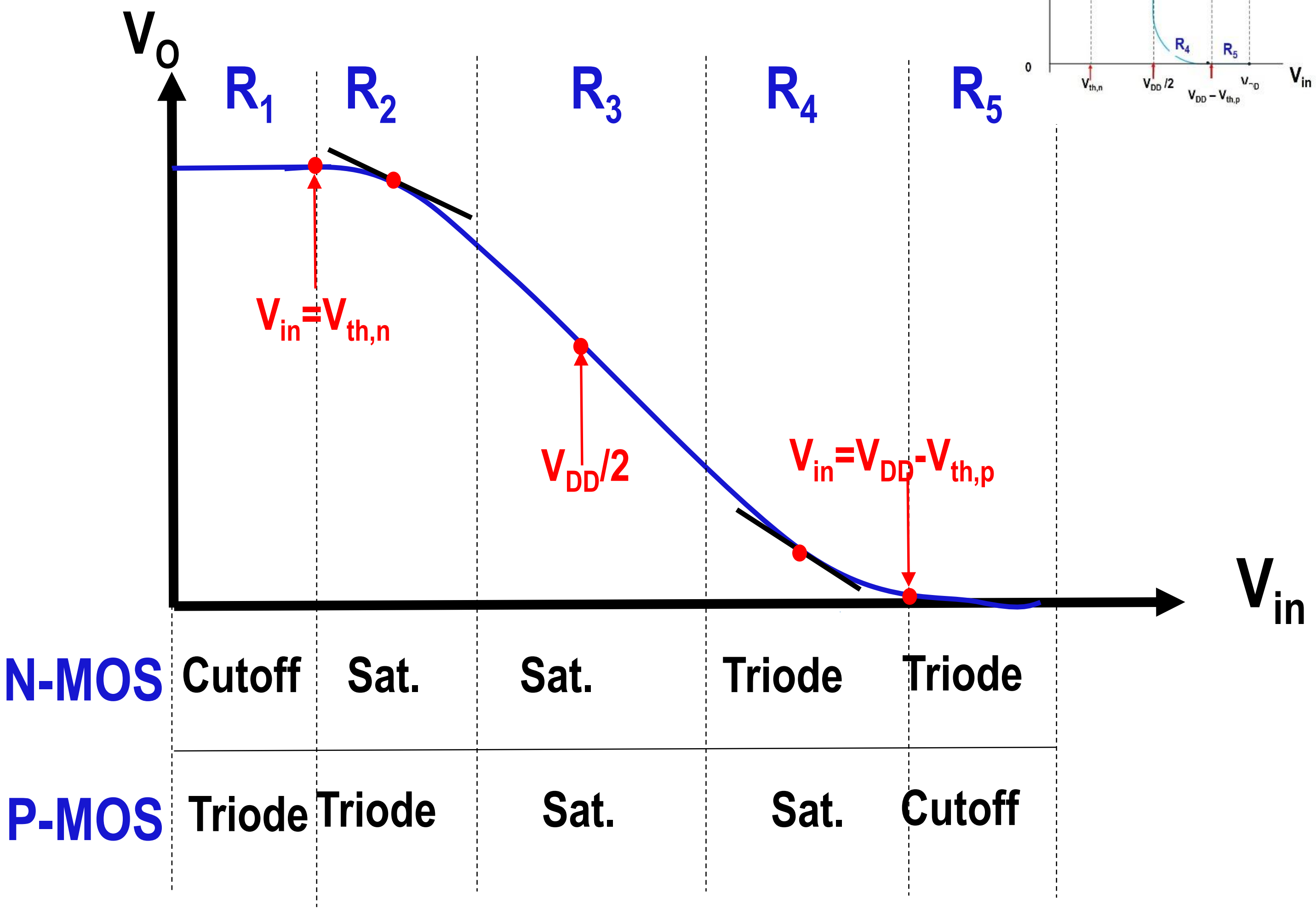


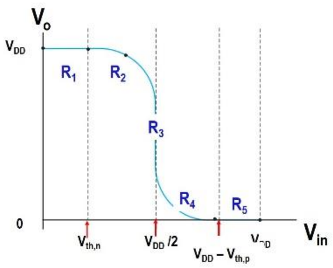


DC Transfer Characteristics (2/8)

- Static CMOS Inverter DC Characteristics



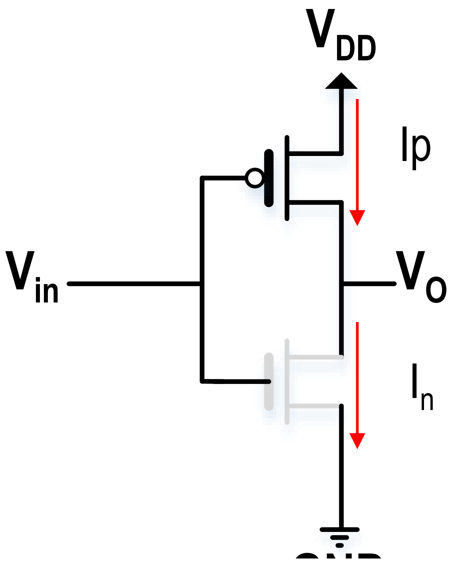


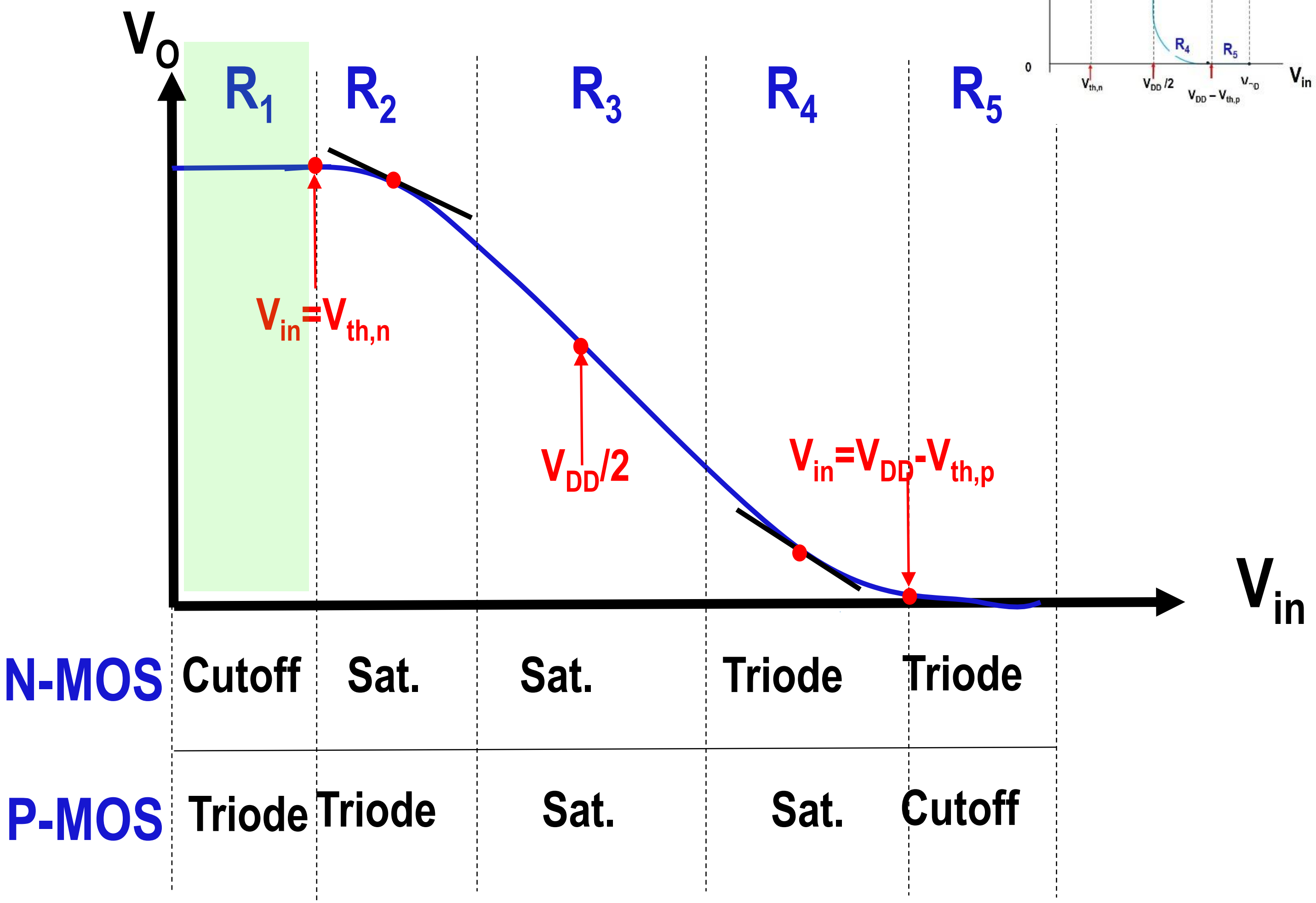


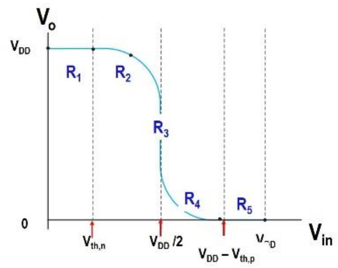
DC Transfer Characteristics (3/8)

❑ Static CMOS Inverter DC Characteristics

Region (1)	
$V_{in} < V_{th,n}$	
N-MOS	**P-MOS**
Cutoff (Open Circuit)	Triode
$V_O = V_{DD}$	
$I_{n(cutoff)} = 0 = I_{p(triode)}$	



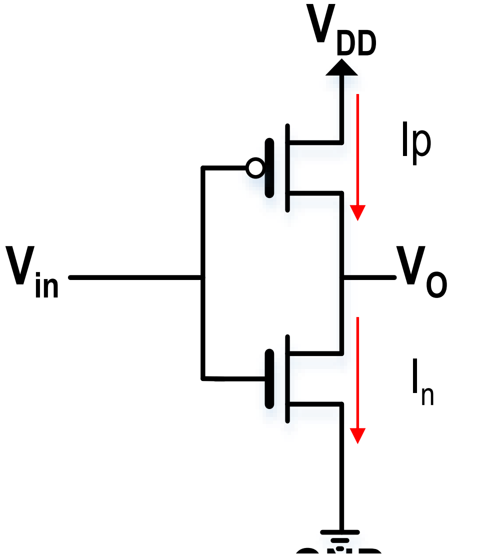


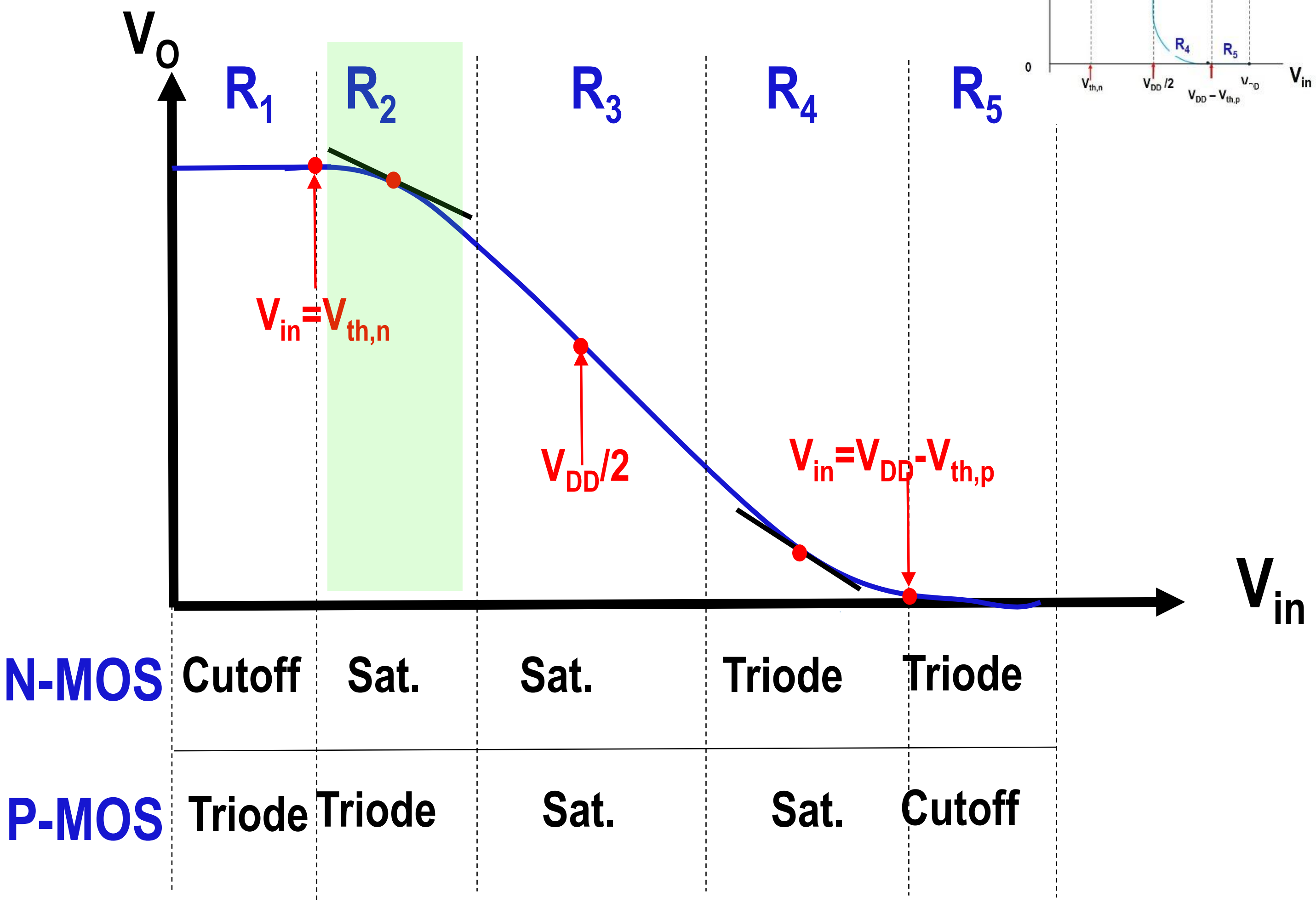


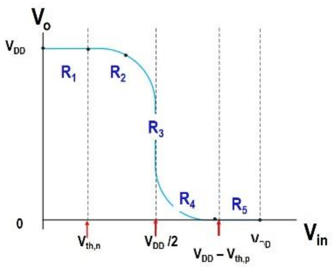
DC Transfer Characteristics (4/8)

❑ Static CMOS Inverter DC Characteristics

Region (2)	
$V_{DD}/2 > V_{in} \geq V_{th,n}$	
N-MOS	**P-MOS**
Saturation	Triode
$V_{ds,n} \geq V_{gs,n} - V_{th,n}$ $V_O \geq V_{in} - V_{th,n}$	$V_{ds,p} > V_{gs,p} - V_{th,p}$ $V_O > V_{in} - V_{th,p}$
$I_{n(Sat.)} = I_{p(triode)}$	



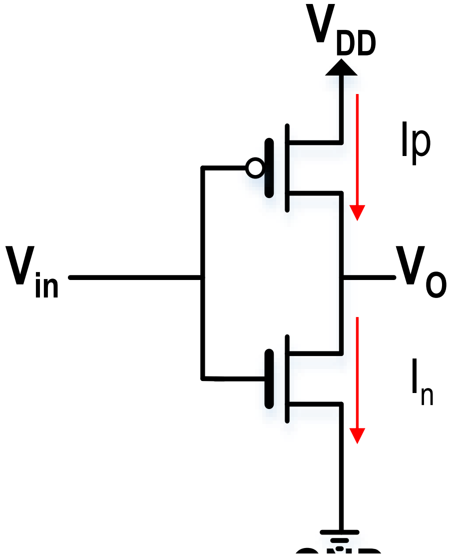


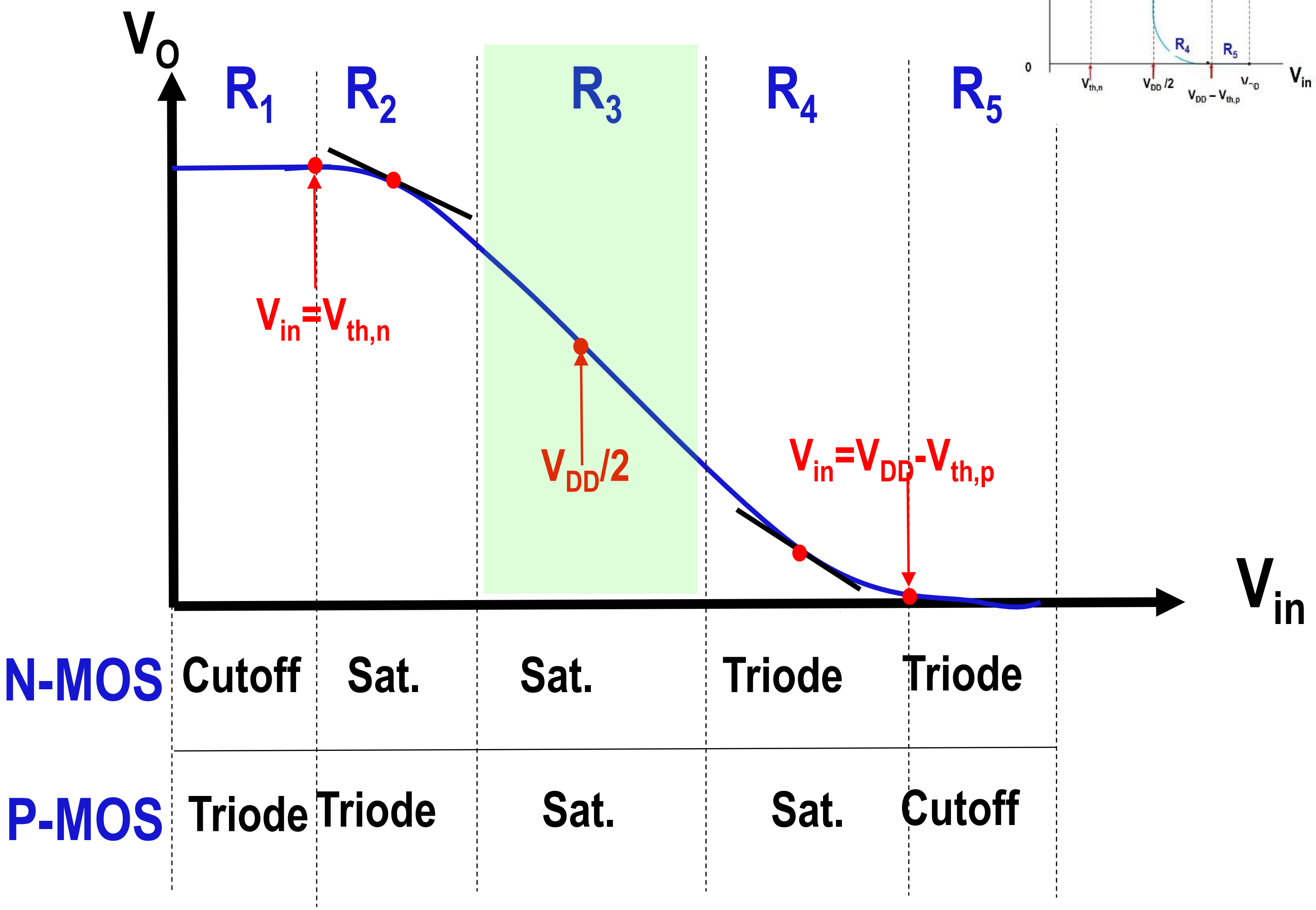


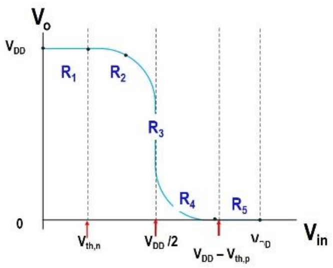
DC Transfer Characteristics (5/8)

❑ Static CMOS Inverter DC Characteristics

Region (3)	
$V_{in} = V_{DD/}2$	
N-MOS	**P-MOS**
Saturation	Saturation
$V_{ds,n} \geq V_{gs,n} - V_{th,n}$ $\mathbf{V_O \geq V_{in} - V_{th,n}}$	$V_{ds,p} \leq V_{gs,p} - V_{th,p}$ $\mathbf{V_O \leq V_{in} - V_{th,p}}$
$I_{n(Sat.)} = I_{p(Sat.)}$	



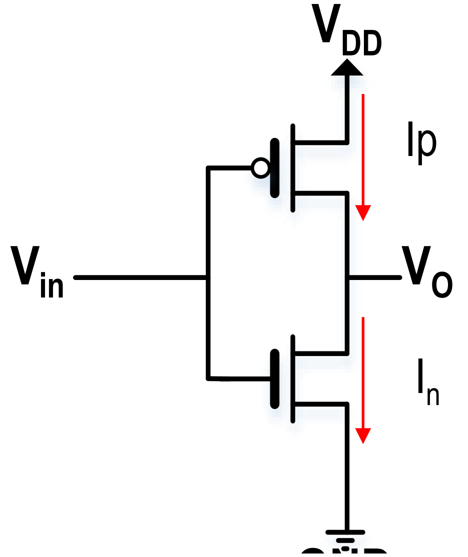


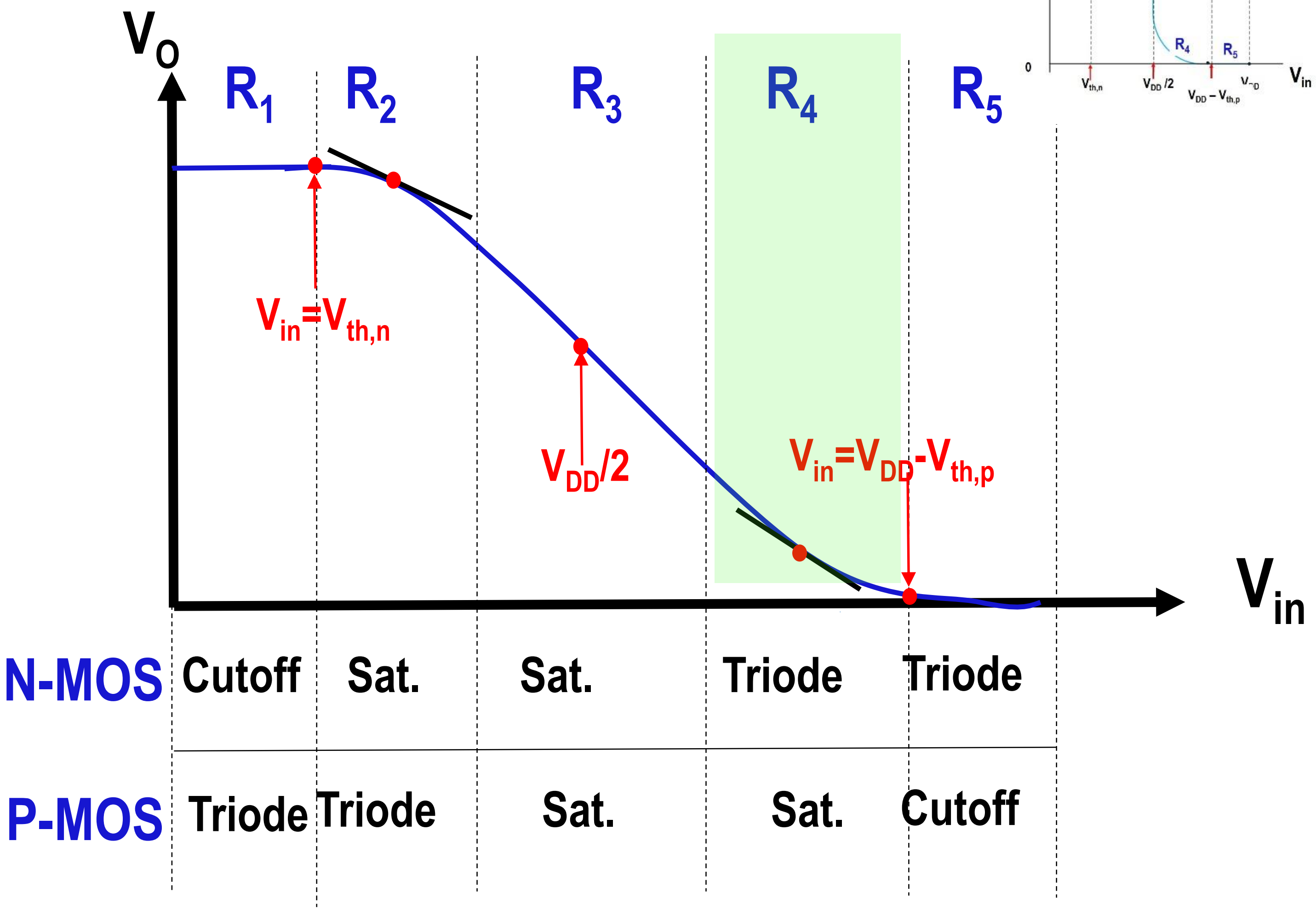


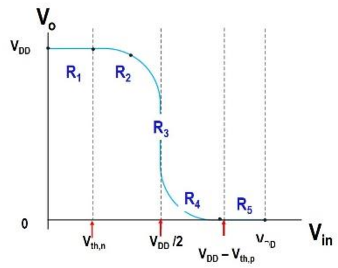
DC Transfer Characteristics (6/8)

❑ Static CMOS Inverter DC Characteristics

Region (4)	
$V_{DD} - V_{th,p} > V_{in} \geq V_{DD}/2$	
N-MOS	**P-MOS**
Triode	Saturation
$V_{ds,n} < V_{gs,n} - V_{th,n}$ $V_O < V_{in} - V_{th,n}$	$V_{ds,p} \leq V_{gs,p} - V_{th,p}$ $V_O \leq V_{in} - V_{th,p}$
$I_{n(Triode)} = I_{p(Sat.)}$	



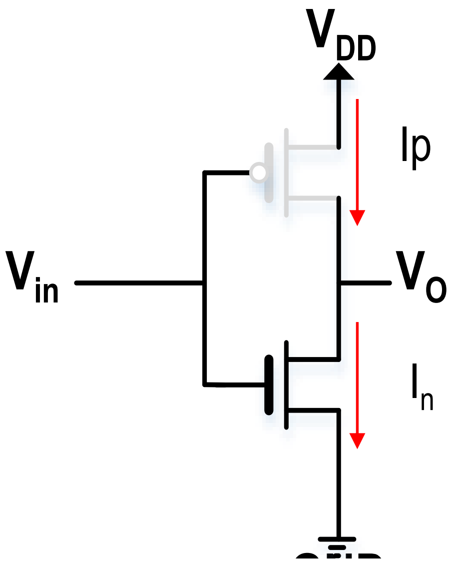


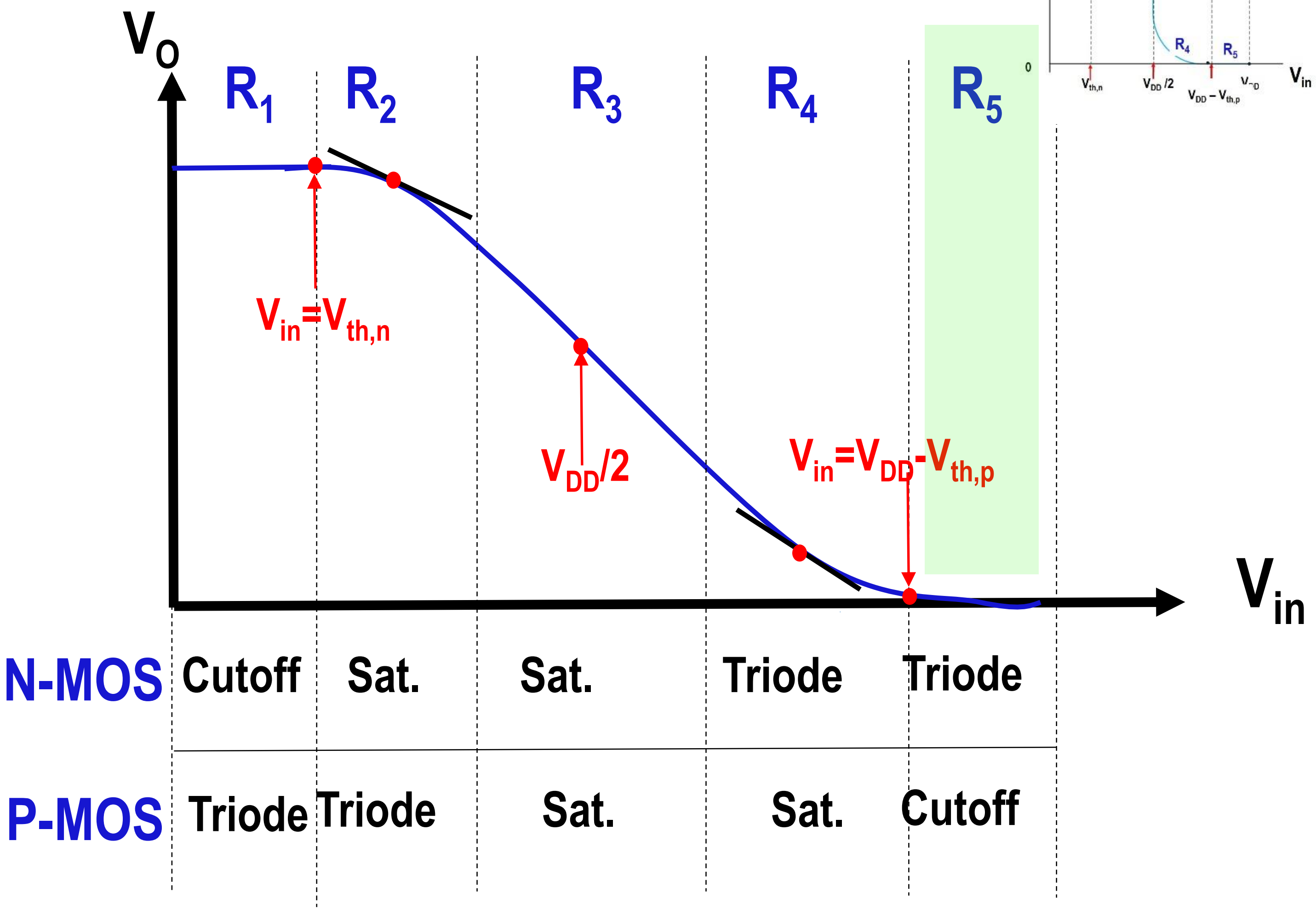


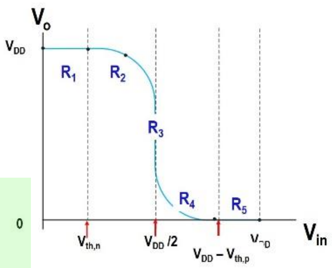
DC Transfer Characteristics (7/8)

❑ Static CMOS Inverter DC Characteristics

Region (5)	
$V_{in} \geq V_{DD} - V_{th,p}$	
N-MOS	**P-MOS**
Triode	Cutoff
$V_O = 0$	
$I_{p(Cutoff)} = 0 = I_{n(Triode)}$	

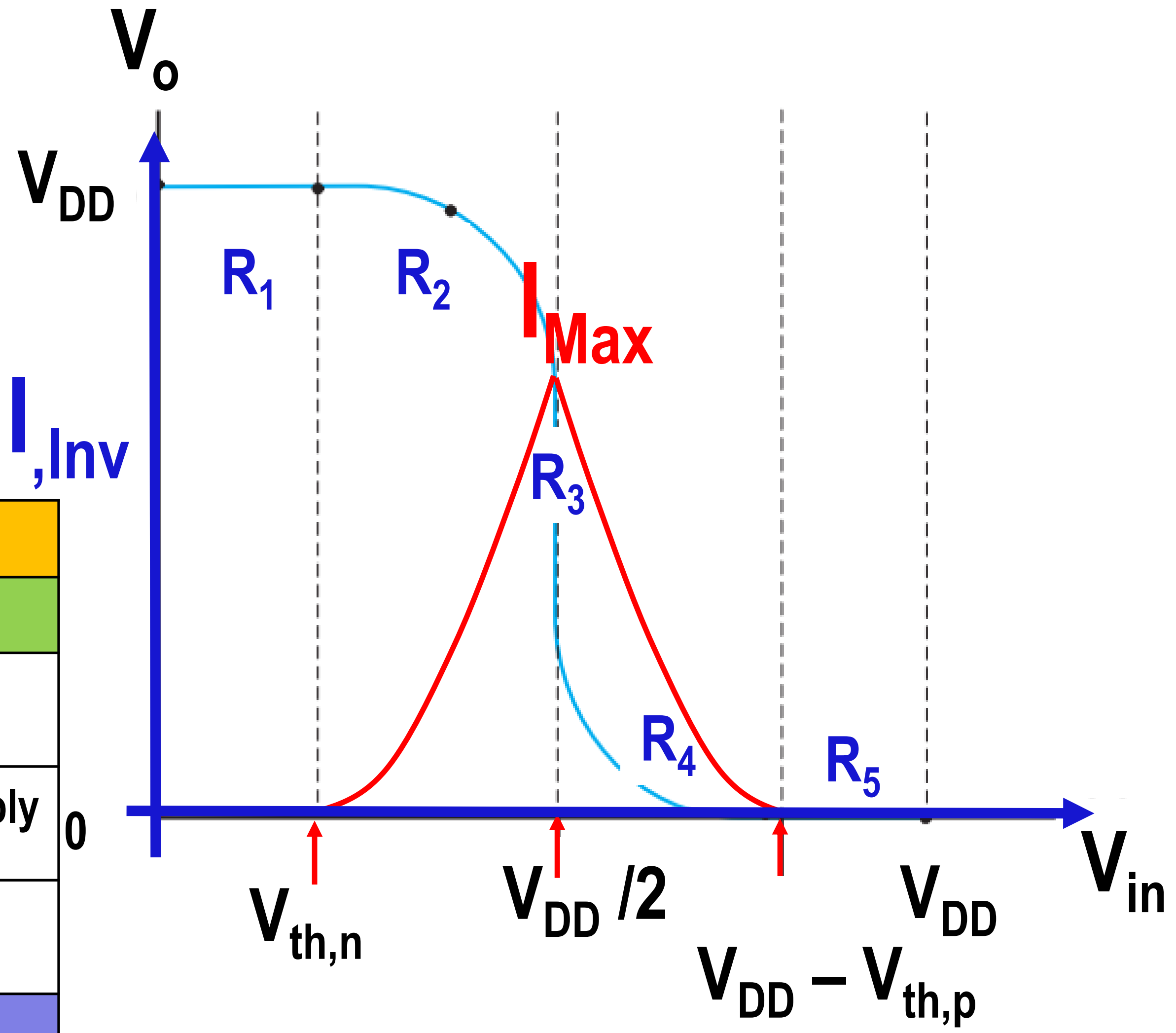






DC Transfer Characteristics (8/8)

- **Maximum** current will be occurred When both N-MOS and P-MOS are operating in saturation region (region 3)



Region	Condition	N-MOS	P-MOS	Output
1	$0<V_{in}<V_{th,n}$	Cutoff	Triode	$V_o=V_{DD}$
2	$V_{th,n} \leq V_{in}<V_{DD}/2$	Sat.	Triode	$V_o>V_{DD}/2$
3	$V_{in}= V_{DD}/2$	Sat.	Sat.	V_o drops sharply
4	$V_{DD}/2 <V_{in} \leq V_{DD}-V_{th,p}$	Triode	Sat.	$V_o <V_{DD}/2$
5	$V_{in} \geq V_{DD}-V_{th,p}$	Triode	Cutoff	$V_o=0$

Outlines

- ❑ DC transfer Ch/CS Curve
- ❑ Noise Margin Concept
- ❑ MOS Inverter Topologies
- ❑ DC Transfer Characteristics
- ❑ Beta Ratio Effects
- ❑ Dynamic Characteristics of CMOS Inverter
- ❑ Inverter switching characteristics
- ❑ Power Dissipation
- ❑ Discussion

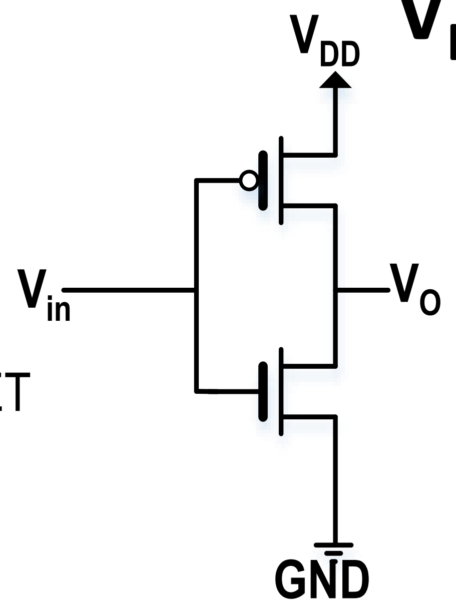
Beta Ratio Effects (1/3)

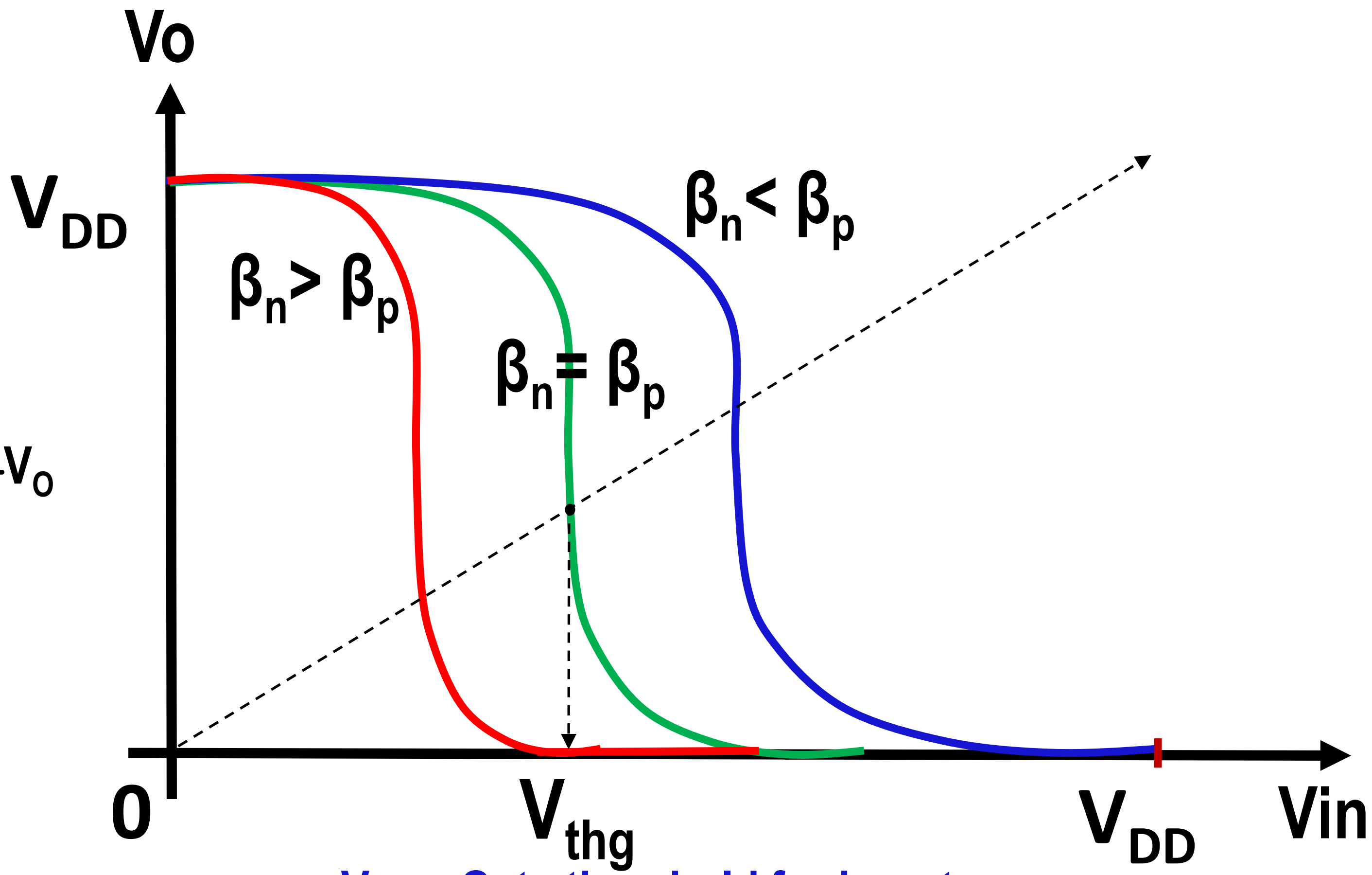
$$B_n = \mu_n C_{ox} \frac{W_n}{L_n}$$

B_n :Gain process factor of n-MOSFET
μ_n: Mobility of Electron
C_{ox}: Gate Oxide Capacitor/unit area
W_n:width of transistor
L_n: length of transistor

$$B_p = \mu_p C_{ox} \frac{W_p}{L_p}$$

B_p :Gain process factor of p-MOSFET
μ_p: Mobility of Hole
C_{ox}: Gate Oxide Capacitor/unit area
W_p: width of transistor of p-MOSFET
L_p: length of transistor of p-MOSFET





V_{thg} : Gate threshold for inverter

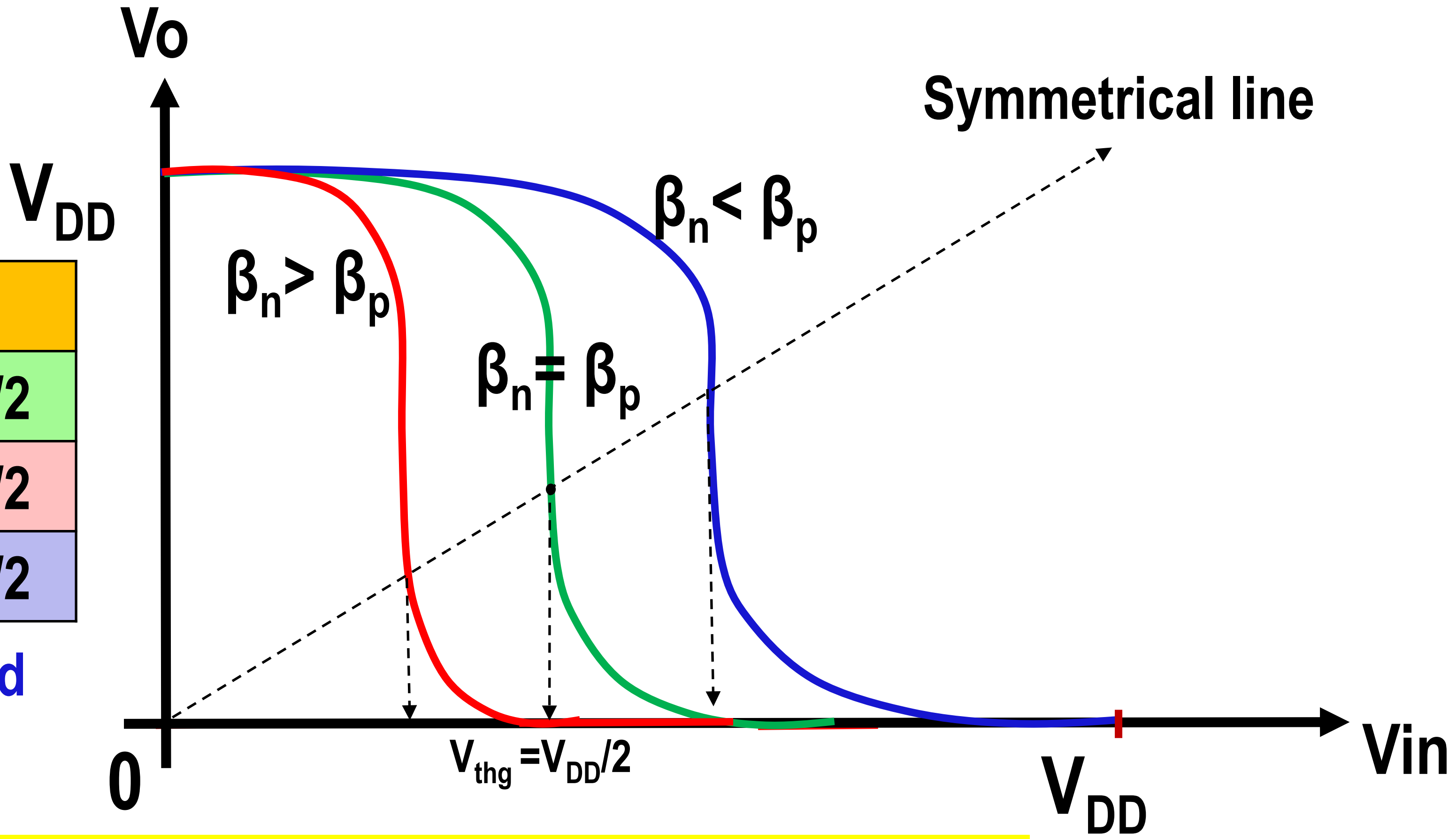
$$V_{thg} = \frac{V_{th,n} + r(V_{DD} + V_{th,p})}{1 + r}$$

Where $r = \sqrt{\frac{\beta_p}{\beta_n}}$

Beta Ratio Effects (2/3)

Case	Condition	V_{thg}
1	$\beta_n = \beta_p$, $V_{th,n} = \lvert V_{th,p} \rvert$	$V_{thg} = V_{DD}/2$
2	$\beta_n > \beta_p$, $V_{th,n} = \lvert V_{th,p} \rvert$	$V_{thg} < V_{DD}/2$
3	$\beta_n < \beta_p$, $V_{th,n} = \lvert V_{th,p} \rvert$	$V_{thg} > V_{DD}/2$

Inverters with different beta are called *skewed* inverters



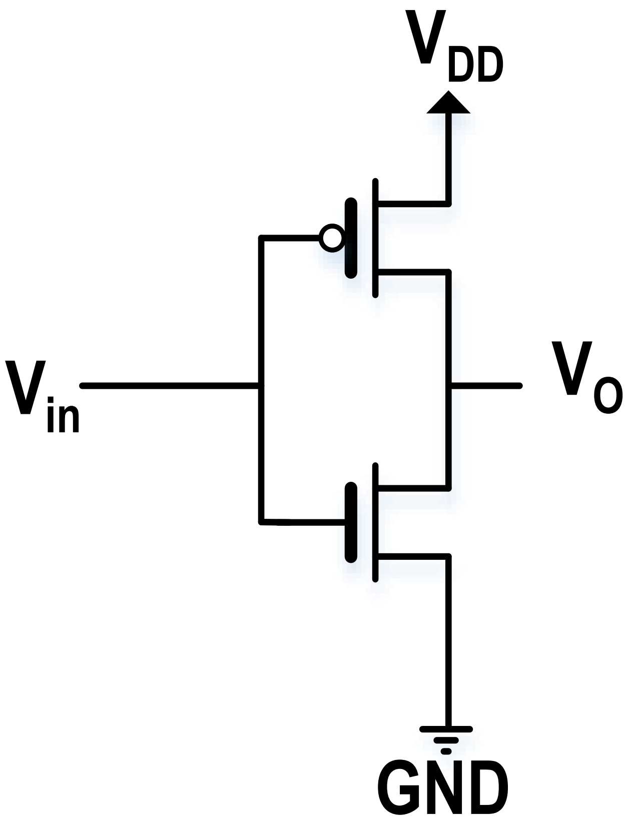
To obtain $\beta_n = \beta_p$, $W_p = 2:3\ W_n$ & $L_p = L_n$ because $\mu_n = 2:3\ \mu_p$

To obtain $\beta_n = \beta_p$, $W_p = 2.5\ W_n$ & $L_p = L_n$ because $\mu_n = 2.5\ \mu_p$

Symmetrical Condition

Beta Ratio Effects (3/3)

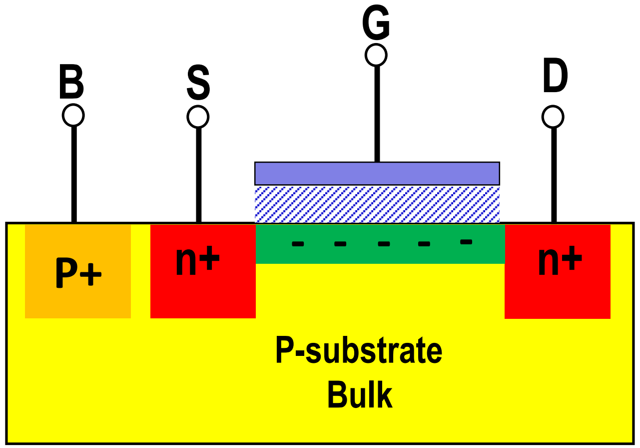
- Symmetrical condition
  - $\beta_n = \beta_p$, $W_p = 2.5\ W_n$ & $L_p = L_n$
- Minimum Silicon Area
  - $\beta_n > \beta_p$, $W_p = W_n$ & $L_p = L_n$



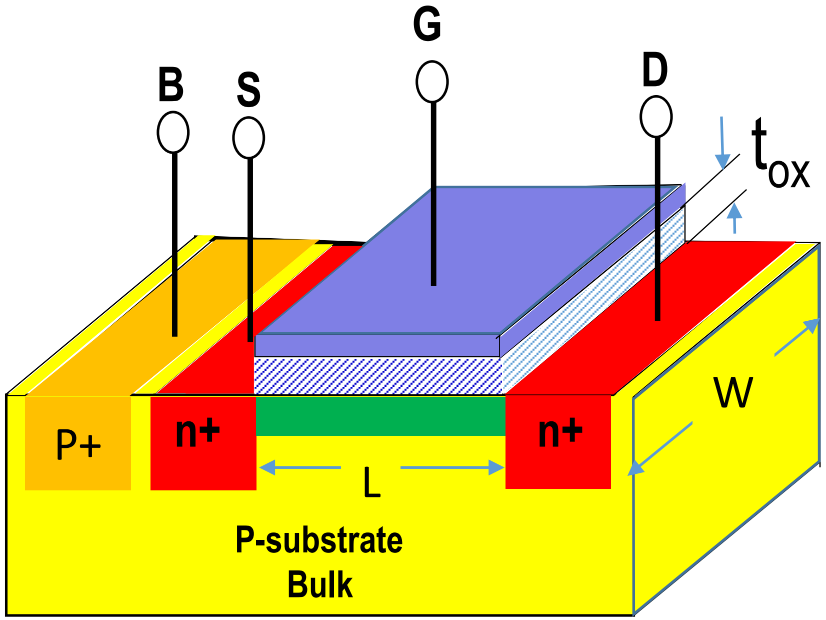
Outlines

- ❑ DC transfer Ch/CS Curve
- ❑ Noise Margin Concept
- ❑ MOS Inverter Topologies
- ❑ DC Transfer Characteristics
- ❑ Beta Ratio Effects
- ❑ Dynamic Characteristics of CMOS Inverter
- ❑ Inverter switching characteristics
- ❑ Power Dissipation
- ❑ Discussion

Dynamic Characteristics of CMOS Inverter (1/2)



2-D n-MOSFET

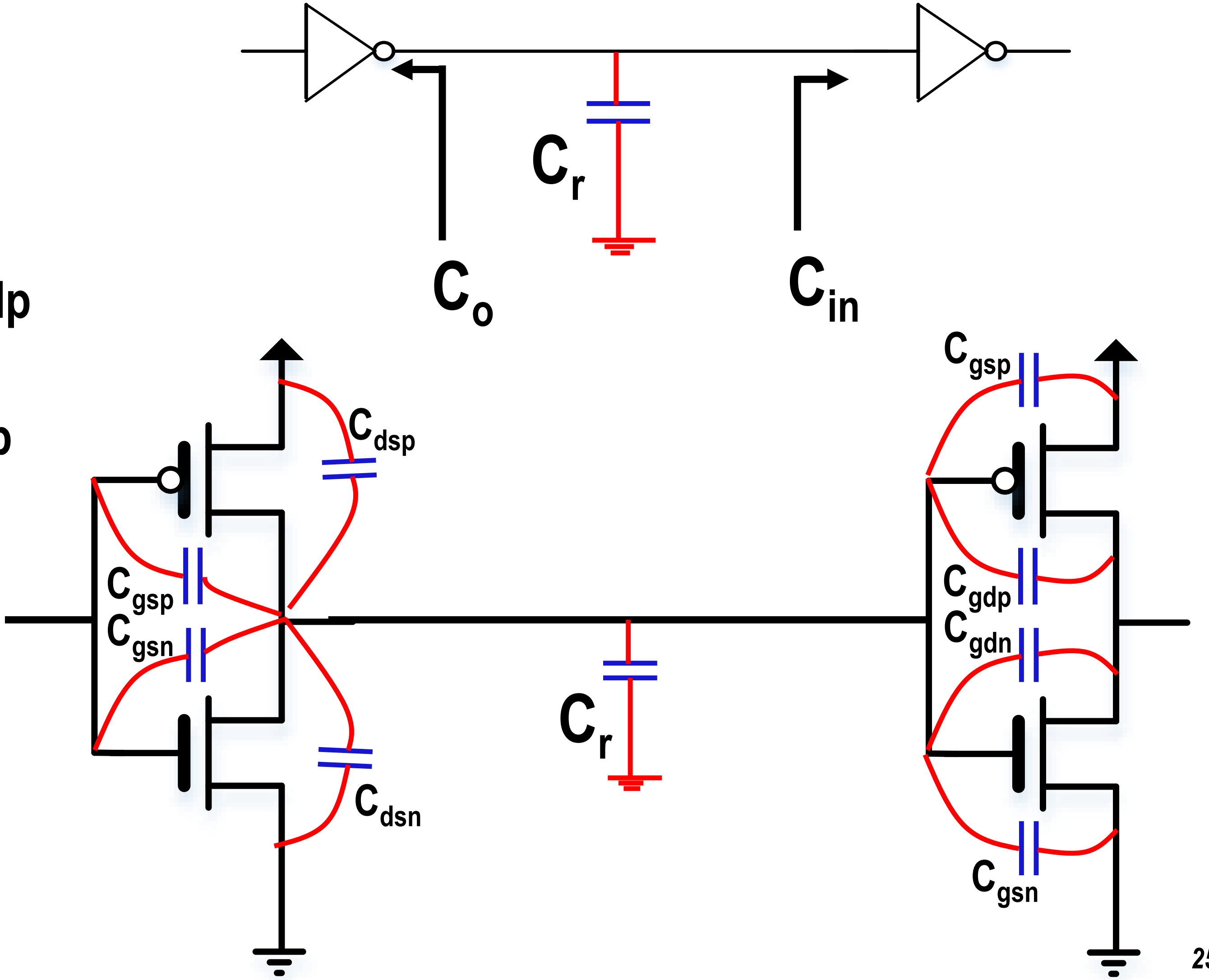


3-D n-MOSFET

The most important element that impacts strongly the dynamic characteristics of CMOS inverter is the capacitor.

❑ Total Capacitance of CMOS Inverter

❑ Assume we have two connected CMOS inverters as shown.



$$C_{total} = C_{in} + C_o + C_r$$

$$C_{in} = C_{gsn} + C_{gsp} + 2^* C_{gdn} + 2^* C_{gdp}$$

$$C_o = C_{dsn} + C_{dsp} + 2^* C_{gdn} + 2^* C_{gdp}$$

C_r : routing capacitance

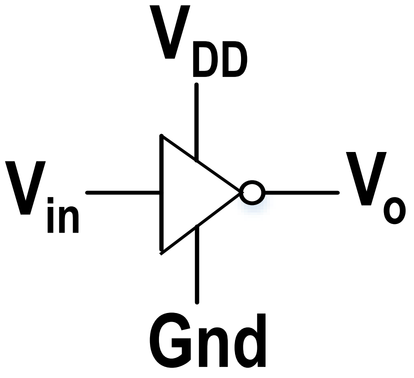
C_o : output capacitance

C_{in} : input capacitance

Outlines

- ❑ DC transfer Ch/CS Curve
- ❑ Noise Margin Concept
- ❑ MOS Inverter Topologies
- ❑ DC Transfer Characteristics
- ❑ Beta Ratio Effects
- ❑ Dynamic Characteristics of CMOS Inverter
- ❑ Inverter switching characteristics
- ❑ Power Dissipation
- ❑ Discussion

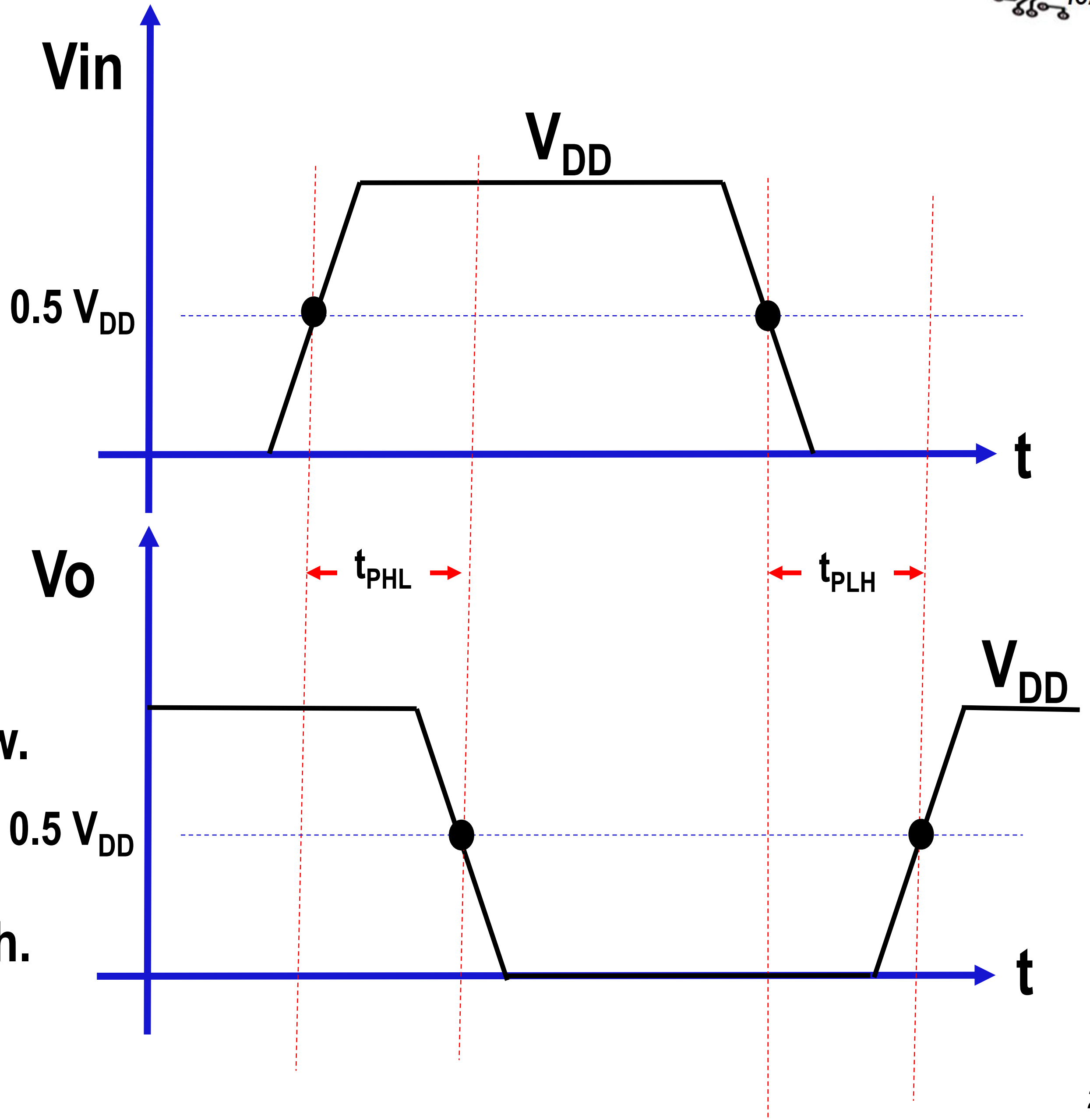
Inverter switching characteristics (1/3)



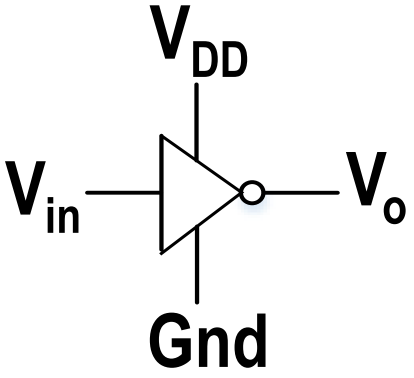
- **Propagation time** (t_p): is the time between 50 % of the input voltage to 50 **%** of the output.

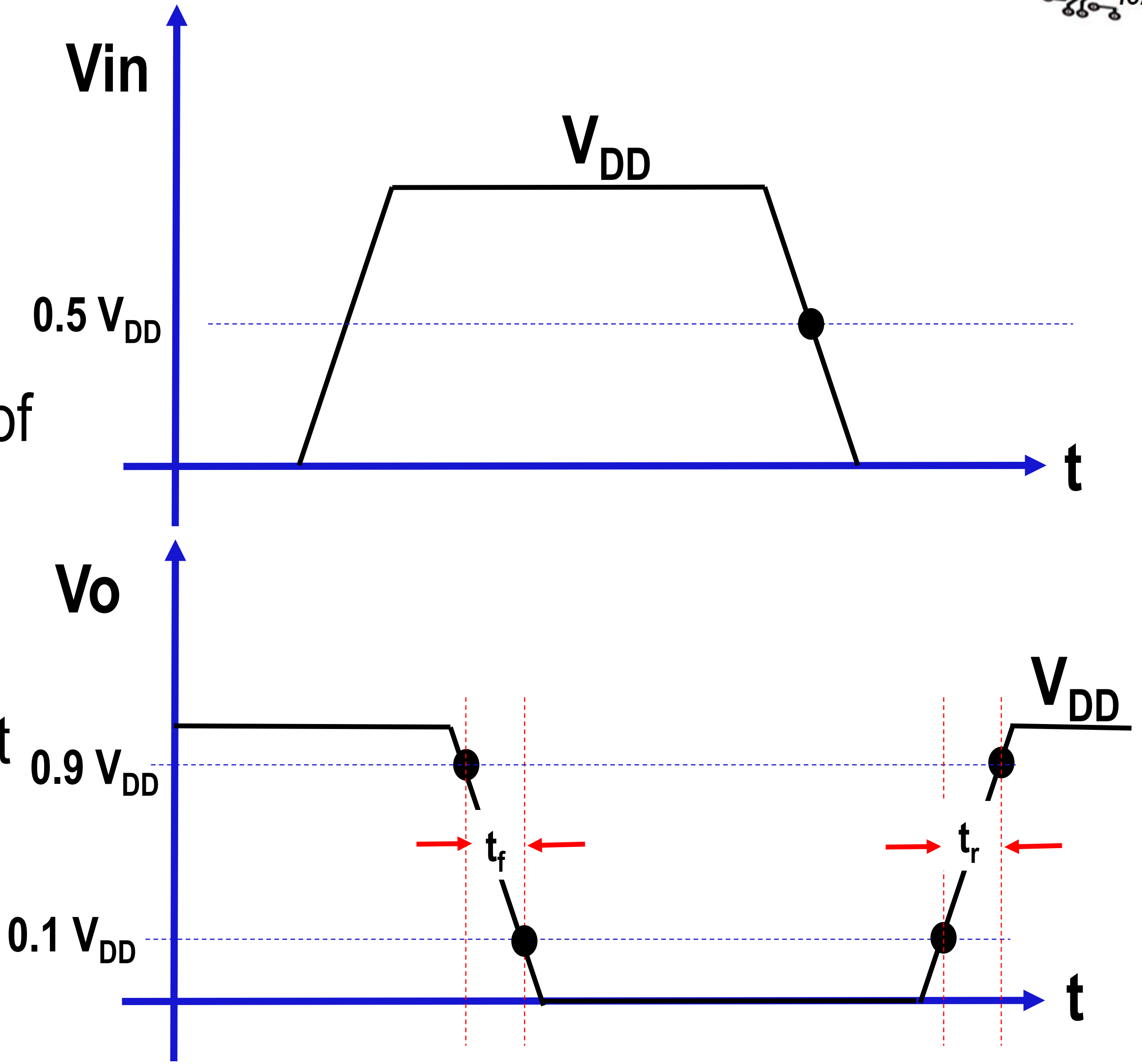
$$t_p = \frac{(t_{PHL} + t_{PLH})}{2}$$

- t_{PHL}: **is the propagation time from high to low.**
- t_{PLH}: **is the propagation time from low to high.**



Inverter switching characteristics (2/3)



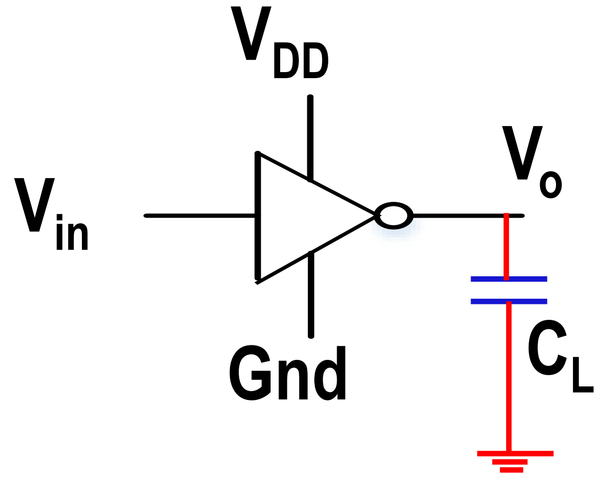


- **Rise time** (**tr**): is the time between 10% of the output voltage to 90 **%** of the output.
- **Fall time** (**tf**):): is the time between 90% of the output voltage to 10**%** of the output

Inverter switching characteristics (3/3)

When, $\beta n = \beta p = \beta$ and $V_{th,n}=V_{th,p}=0.2*V_{DD}$

$$\mathbf{t_p} = \frac{1.61 * C_L}{\beta * V_{DD}}$$



$$\mathbf{t_r} = t_f = \frac{3.67 * C_L}{\beta * V_{DD}}$$

Propagation time, rising time and falling time are dependent on

- ❑Load capacitance C_L
- ❑Power supply VDD
- ❑Gain process factor β

Outlines

- ❑ DC transfer Ch/CS Curve
- ❑ Noise Margin Concept
- ❑ MOS Inverter Topologies
- ❑ DC Transfer Characteristics
- ❑ Beta Ratio Effects
- ❑ Dynamic Characteristics of CMOS Inverter
- ❑ Inverter switching characteristics
- ❑ Power Dissipation
- ❑ Discussion

Power Dissipation (1/4)

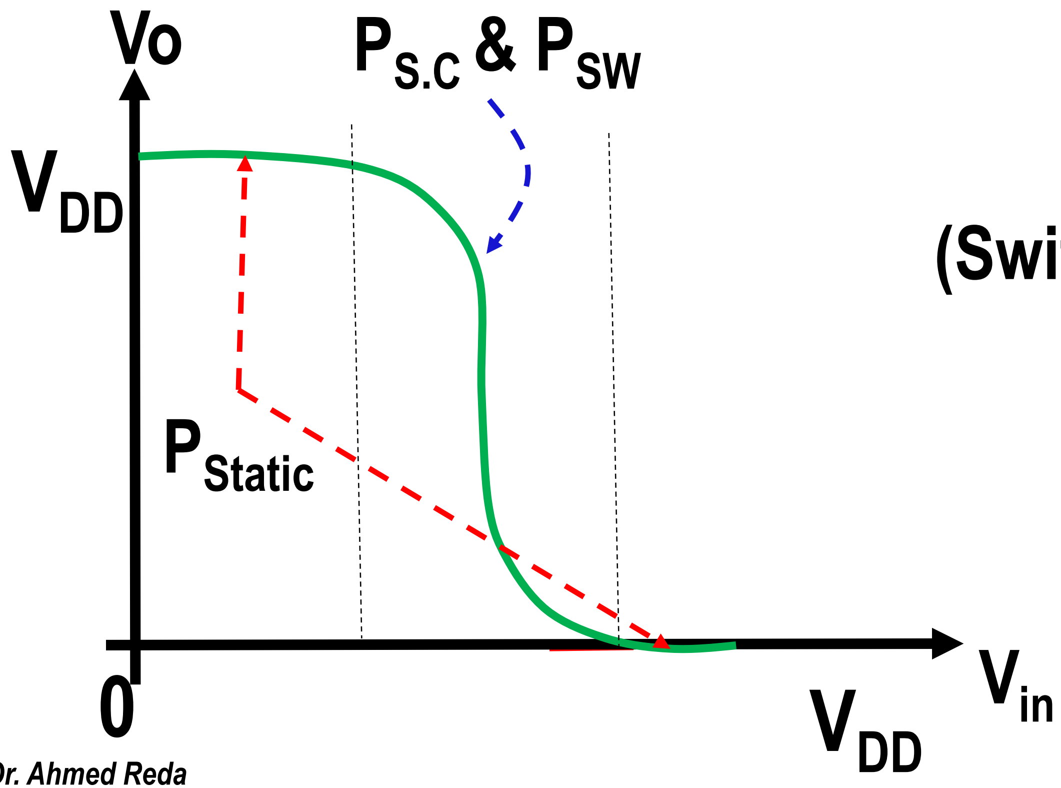
Power Dissipation

Static power dissipation (P_{static})

Dynamic power dissipation

Charging/Discharging (Switching) Power Dissipation (P_{SW})

Short Circuit Power Dissipation ($P_{S.C}$)



$$P_{Total} = P_{Static} + P_{Dyamic}$$

$$P_{Total} = P_{Static} + P_{Sw} + P_{S.C}$$

Static Power Dissipation

- **Definition:**

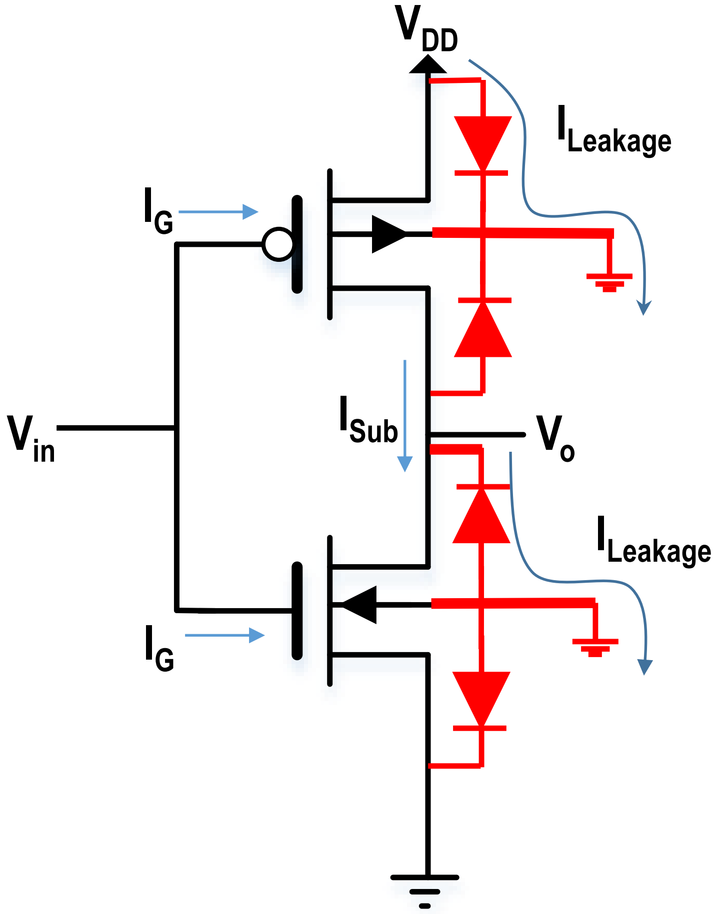
Is the consumed power when passing a constant current due to apply a constant voltage.

- **Source of $P_{static.}$**
  - **Sub-threshold leakage**

    When $V_{gs,n} < V_{th,n}$ → there is a very small current ($I_{leakage}$) in range of Pico-Ampere will pass through the transistor
  - **Gate-leakage**

    Current would pass from gate to the channel→. I_G=0
  - **Junction-leakage**

    Current would pass through the parasitic diodes due to junction



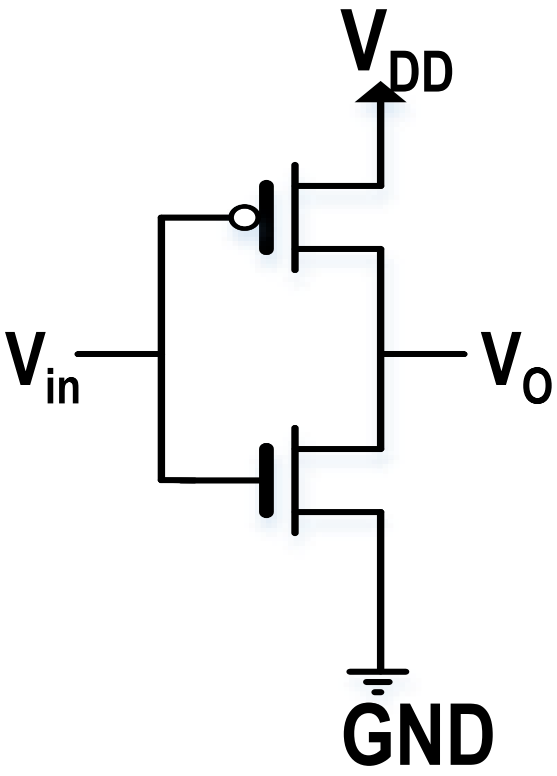
$$P_{Static} = (I_{sub}+I_G+I_{Junction})*V_{DD}$$

Power Dissipation (3/4)

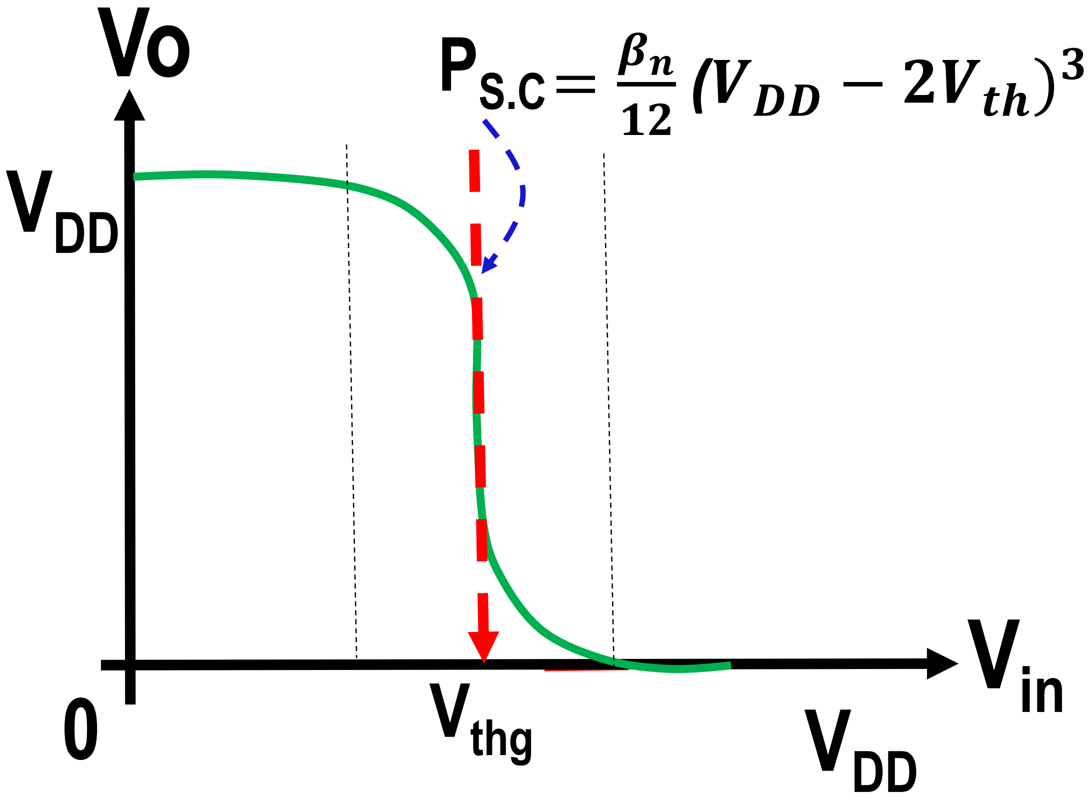
Shot circuit power dissipation

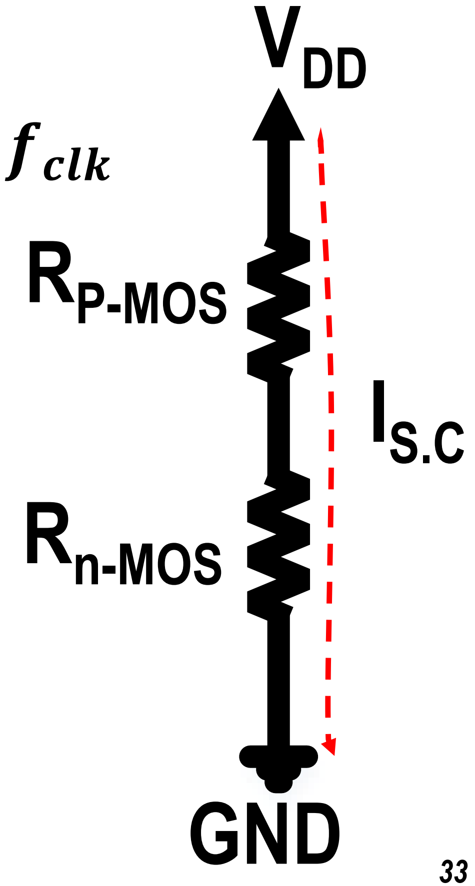
❑ Definition:

Is the consumed power within the transient region. It is called power of short circuit because the current will pass from V_{DD} to GND while the N-MOS and P-MOS are operating “ON”.



$$P_{S.C} = \frac{\beta_n}{12}(V_{DD} - 2V_{th})^3 * t_r * f_{clk}$$



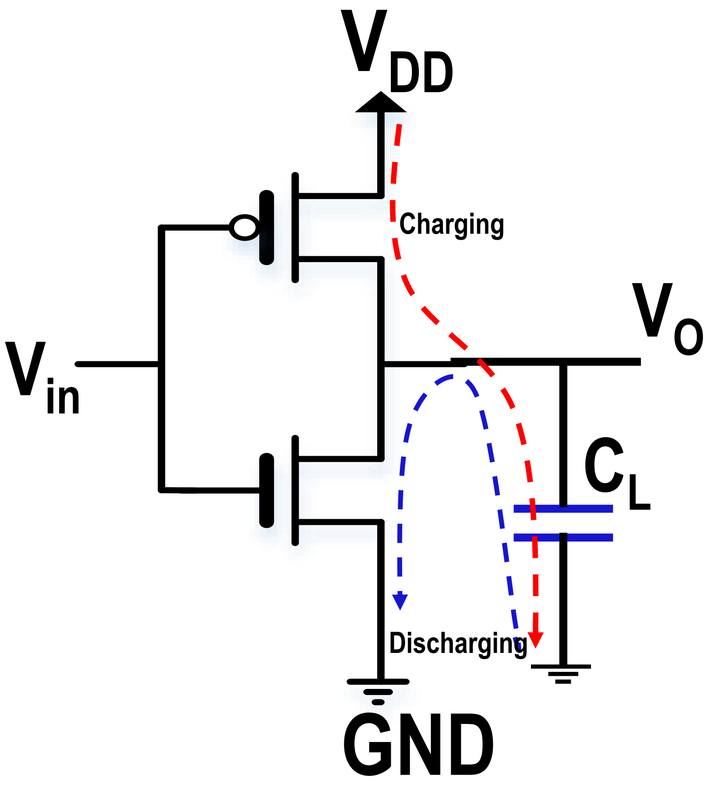


Power Dissipation (4/4)

Switching Power Dissipation "Charging/discharging"

- **Definition:**

Is the consumed power during charging and discharging the capacitor



$$P_{SW} = C_L * V_{DD}^2 * f_{sw}$$
$$= \alpha * C_L * V_{DD}^2 * f_{clk}$$

f_{sw} : Switching frequency

f_{clk} : System clock frequency

$f_{sw} = \alpha * f_{clk}$

α: Activity factor ($0 < \alpha < 1$)

Note

$P_{SW} >>> P_{S.C}$

Outlines

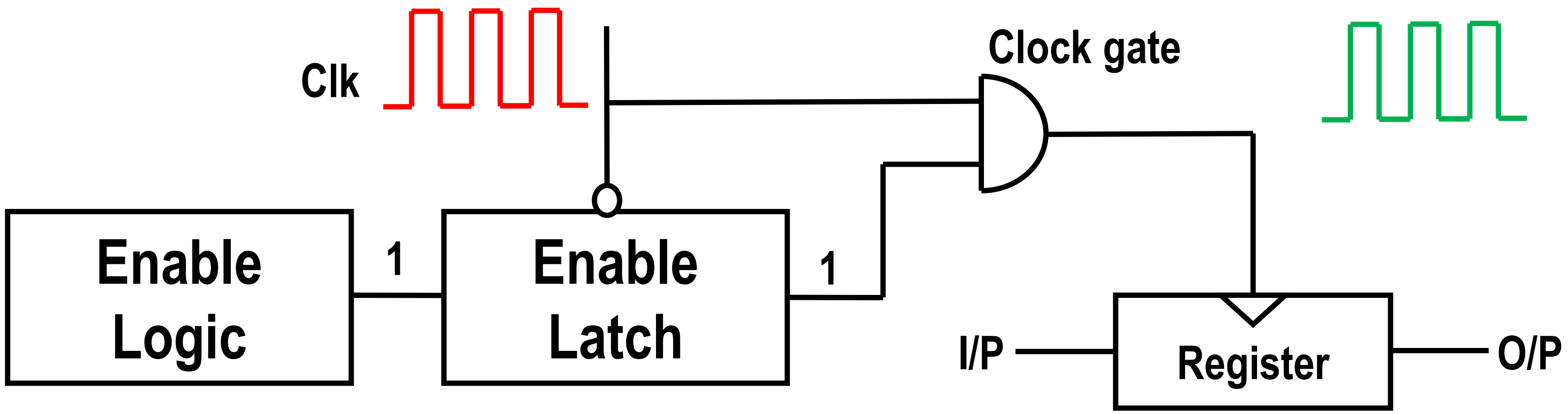
- ❑ DC transfer Ch/CS Curve
- ❑ Noise Margin Concept
- ❑ MOS Inverter Topologies
- ❑ DC Transfer Characteristics
- ❑ Beta Ratio Effects
- ❑ Dynamic Characteristics of CMOS Inverter
- ❑ Inverter switching characteristics
- ❑ Power Dissipation
- ❑ Discussion

Discussion (1/4)

- What are the dynamic power reduction factors?
  - Capacitance (C)
  - Supply Voltage (V_{DD})
  - Clock frequency (F_{clk})
  - Activity factor (α)

$$P_{SW} = \alpha * C_L * V_{DD}^2 * f_{clk}$$

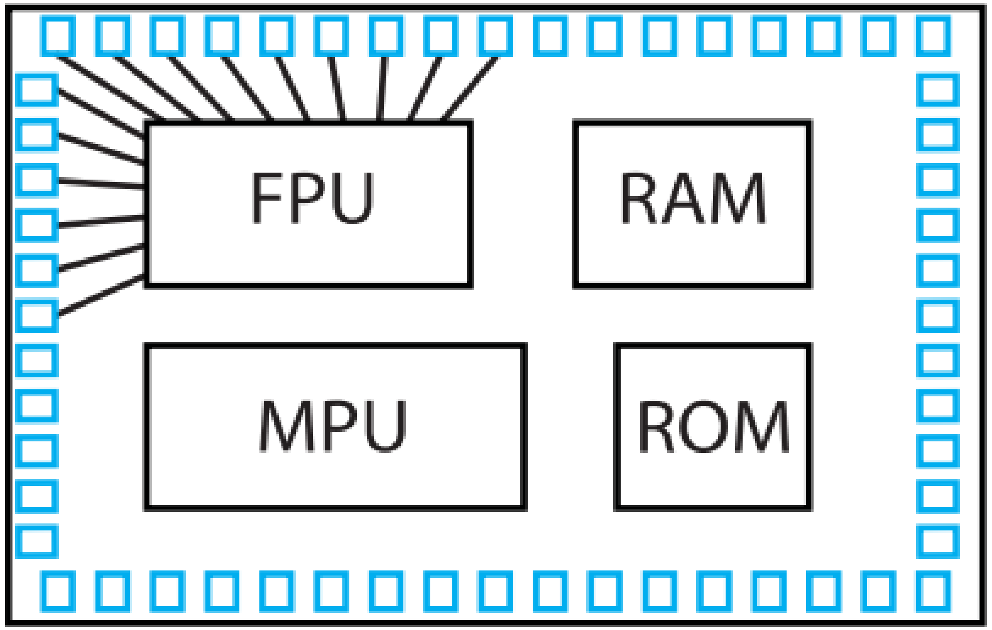
- How you can reduce the activity factor?
  - The proper method is tuning off the clock which drives the un-used register.



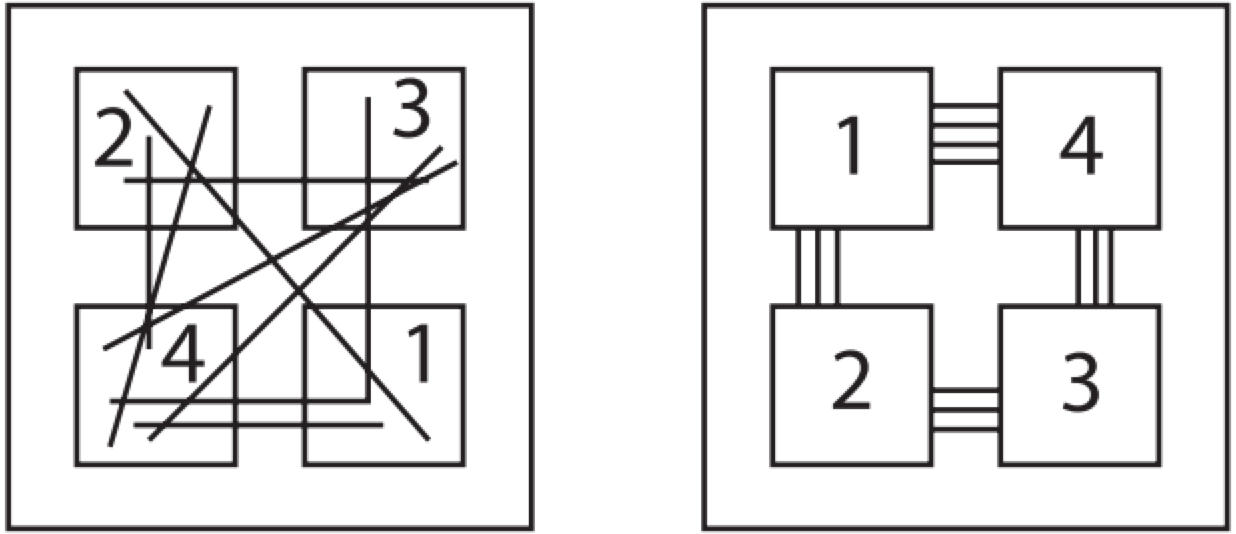
Discussion (2/4)

- How you can reduce the capacitance of the circuit?
- There are two existing methods to reduce the circuit's capacitances
  - Gate's capacitance
    - Use fewer stages of logic gate
    - Use small size of the gate (W, L)
  - Wire capacitance
    - Good floor planning is placement of the electronic elements inside the integrated circuit to keep communication blocks close to each other. Thus, the resulting parasitic capacitance will be reduced strongly.
    - If we had to use long wires, we should use inverters or buffers for driving.

Discussion (3/4)



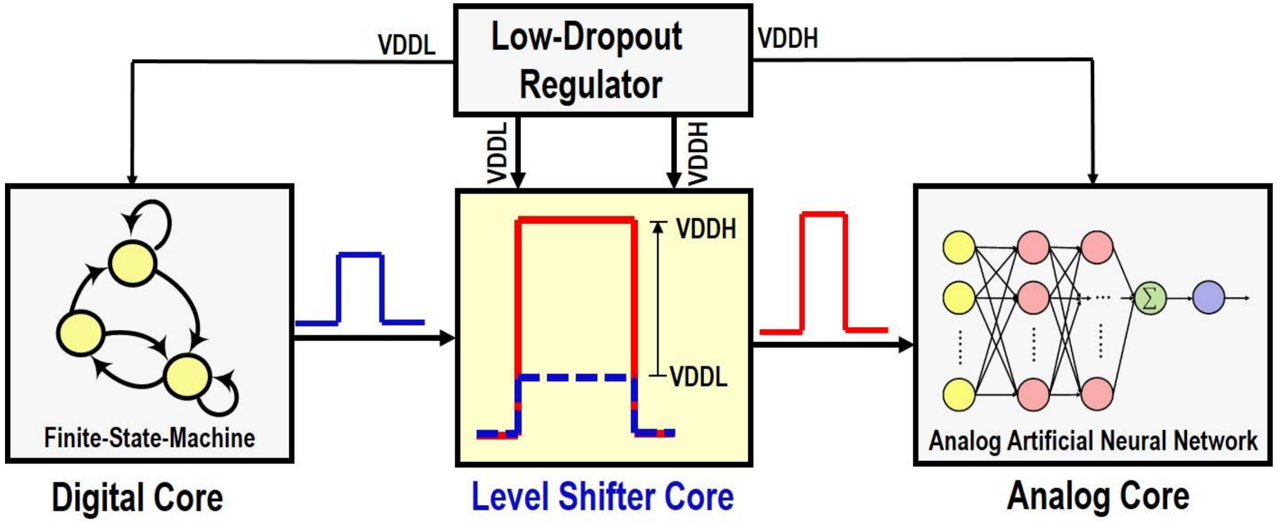
Inputs and outputs can be located near their appropriate cell block



Neat flylines indicate good floor-planning

Discussion (4/4)

- ❑ How you organize the voltage domain (Supply voltage) to reduce the power dissipation?
  - ❑ You can provide separated supply voltage to different blocks
  - ❑ Sometimes, a low power supply can be used for digital core while the analog core use higher supply voltage.
  - ❑ Usually, a level shifter with up/down conversion voltage can be used.



Conceptual Mixed-Mode System-On-Chip

Thanks!